HIKROBOT

RCS-2000 API

Developer Guide

Legal Information

© Hangzhou Hikrobot Co., Ltd. All Rights Reserved.

About this Document

The Document includes instructions for using and managing the Product. Pictures, charts, images and all other information hereinafter are for description and explanation only. The information contained in the Document is subject to change, without notice, due to firmware updates or other reasons. Please find the latest version in the company website (***https://en.hikrobotics.com***). Unless otherwise agreed, Hangzhou Hikrobot Co., Ltd. or its affiliates (hereinafter referred to as “Hikrobot”) makes no warranties, express or implied.
Please use this Document with the guidance and assistance of professionals trained in supporting the Product.

Acknowledgment of Intellectual Property Rights

- Hikrobot owns the copyrights and/or patents related to the technology embodied in the Products described in this Document, which may include licenses obtained from third parties. Any part of the Document, including text, pictures, graphics, etc., belongs to Hikrobot. No part of this Document may be excerpted, copied, translated, or modified in whole or in part by any means without written permission.
- ***HIKROBOT*** and other Hikrobot's trademarks and logos are the properties of Hikrobot in various jurisdictions. Other trademarks and logos mentioned are the properties of their respective owners.

LEGAL DISCLAIMER

TO THE MAXIMUM EXTENT PERMITTED BY APPLICABLE LAW, THIS MANUAL AND THE PRODUCT DESCRIBED, WITH ITS HARDWARE, SOFTWARE AND FIRMWARE, ARE PROVIDED "AS IS" AND "WITH ALL FAULTS AND ERRORS". HIKROBOT MAKES NO WARRANTIES, EXPRESS OR IMPLIED, INCLUDING WITHOUT LIMITATION, MERCHANTABILITY, SATISFACTORY QUALITY, OR FITNESS FOR A PARTICULAR PURPOSE. THE USE OF THE PRODUCT BY YOU IS AT YOUR OWN RISK. IN NO EVENT WILL HIKROBOT BE LIABLE TO YOU FOR ANY SPECIAL, CONSEQUENTIAL, INCIDENTAL, OR INDIRECT DAMAGES, INCLUDING, AMONG OTHERS, DAMAGES FOR LOSS OF BUSINESS PROFITS, BUSINESS INTERRUPTION, OR LOSS OF DATA, CORRUPTION OF SYSTEMS, OR LOSS OF DOCUMENTATION, WHETHER BASED ON BREACH OF CONTRACT, TORT (INCLUDING NEGLIGENCE), PRODUCT LIABILITY, OR OTHERWISE, IN CONNECTION WITH THE USE OF THE PRODUCT, EVEN IF HIKROBOT HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGES OR LOSS.
YOU ACKNOWLEDGE THAT THE NATURE OF INTERNET PROVIDES FOR INHERENT SECURITY RISKS, AND HIKROBOT SHALL NOT TAKE ANY RESPONSIBILITIES FOR ABNORMAL OPERATION, PRIVACY LEAKAGE OR OTHER DAMAGES RESULTING FROM

CYBER-ATTACK, HACKER ATTACK, VIRUS INFECTION, OR OTHER INTERNET SECURITY RISKS; HOWEVER, HIKROBOT WILL PROVIDE TIMELY TECHNICAL SUPPORT IF REQUIRED.
YOU AGREE TO USE THIS PRODUCT IN COMPLIANCE WITH ALL APPLICABLE LAWS, AND YOU ARE SOLELY RESPONSIBLE FOR ENSURING THAT YOUR USE CONFORMS TO THE APPLICABLE LAW. ESPECIALLY, YOU ARE RESPONSIBLE, FOR USING THIS PRODUCT IN A MANNER THAT DOES NOT INFRINGE ON THE RIGHTS OF THIRD PARTIES, INCLUDING WITHOUT LIMITATION, RIGHTS OF PUBLICITY, INTELLECTUAL PROPERTY RIGHTS, OR DATA PROTECTION AND OTHER PRIVACY RIGHTS. YOU SHALL NOT USE THIS PRODUCT FOR ANY PROHIBITED END-USES, INCLUDING THE DEVELOPMENT OR PRODUCTION OF WEAPONS OF MASS DESTRUCTION, THE DEVELOPMENT OR PRODUCTION OF CHEMICAL OR BIOLOGICAL WEAPONS, ANY ACTIVITIES IN THE CONTEXT RELATED TO ANY NUCLEAR EXPLOSIVE OR UNSAFE NUCLEAR FUEL-CYCLE, OR IN SUPPORT OF HUMAN RIGHTS ABUSES.
THE PERFORMANCE DATA IN THIS PUBLICATION IS BASED ON HIKROBOT'S INTERNAL RESEARCH/EVALUATION. ACTUAL DATA MAY VARY DEPENDING ON SPECIFIC CONFIGURATIONS AND OPERATING CONDITIONS AND HIKROBOT SHALL NOT BEAR THE CONSEQUENCES ARISING THEREFROM.
IN THE EVENT OF ANY CONFLICTS BETWEEN THIS MANUAL AND THE APPLICABLE LAW, THE LATTER PREVAILS.

TABLE OF CONTENTS

Chapter 1 Overview 1

1.1 Request Header Fields 1

1.2 Request Message Body 2

1.3 Encrypted Signature 2

1.4 Response Header Fields 5

1.5 Response Message Body 6

1.6 Response Status Codes 7

1.7 General Codes for Response Message Body 8

1.8 Update History 8

Chapter 2 API Reference 14

2.1 AMR Scheduling APIs 14

2.1.1 Task Group 14

2.1.2 Apply Task 18

2.1.3 Continue Task 27

2.1.4 Cancel Task 34

2.1.5 Set Task Priority 39

2.1.6 Pause and Restore AMRs in Specified Area 42

2.1.7 Return AMR to Fixed Area 46

2.1.8 Clear Area 50

2.1.9 Block and Release Area 53

2.1.10 Link Carrier to Point 57

2.1.11 Unlink Carrier from Point 59

2.1.12 Link or Unlink Storage Object to or from Carrier 62

2.1.13 Enable or Disable Carrier 65

2.1.14 Enable or Disable Point 68

2.1.15 Notify of Third-Party Device Execution (Return Code: 0) 71

2.1.16 Apply Pre-Scheduling Task 74

2.1.17 Search Task Status 77
2.1.18 Search AMR Status 83
2.1.19 Search Carrier Status 89
2.1.20 Link Carrier to Material....92
2.1.21 Unlink Carrier from Material 95
2.1.22 Notify of Third-Party Device Execution (Return Code: SUCCESS)....97
2.2 SPI APIs for Receiving Feedback 101
2.2.1 Receive Feedback on Task Performance 101
2.2.2 Request for Resources....106
2.2.3 Request for Third-Party Device....110
2.2.4 Receive Feedback on Completion of AMR Returning....114
2.2.5 Receive Feedback on Completion of Area Clearing 116
2.2.6 Report AMR Exception Alarm 119
2.2.7 Report Task Exception Alarm 121
2.2.8 Notify of Linking/Unlinking 124
2.2.9 Verify Tote Inbound at FlashStation 129
Chapter 3 Samples for Calling Task Applying API 132
3.1 LMR General Transportation Task....132
3.2 LMR Task – Initialization of Specified Carrier 132
3.3 Carrier Detection and Transportation 134
3.4 CTU General Transportation Task 135
3.5 CTU Workstation Inbound....136
3.6 CTU Task – Buffer Rack Outbound....137
3.7 CTU Task – Buffer Rack Inbound....138
3.8 CTU Conveyor Detection Inbound 138
3.9 CTU Conveyor Inbound 139
3.10 FMR General Transportation Task....140
3.11 Roller CMR General Transportation Task....141

Chapter 4 Typical Scheduling Scenes 142

Chapter 1 Overview

When the RCS-2000 calls the APIs provided by the third-party platform, the default connection timeout threshold is 30 seconds, and the default returning timeout threshold is 60 seconds. If the timeout is longer than the default thresholds, the RCS-2000 will return the information of connection failed.

If connecting to the third-party platform failed, you can try again after five seconds, and by default, up to 5 failed connection attempts are allowed.

1.1 Request Header Fields

Default request header fields are detailed below.

Field Name	Data Type	Max. Length	Req. or Opt.	Description
Authorization	String	N/A	Opt.	Fixed format: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
Content-Type	String	N/A	Opt.	Fixed value: application/json;charset=UTF-8
X-lr-appkey	String	32	Opt.	Unique ID issued to business system by scheduling system.
X-lr-request-id	String	16	Req.	Unique request ID.
X-lr-version	String	12	Opt.	API version. To ensure backward compatibility, API versioning is adopted.
X-lr-trace-id	String	32	Opt.	Full-stack tracing ID. It is used to identify upstream and downstream faults, and it should be returned in the response result. UUID is recommended.
X-lr-source	String	32	Opt.	Command source for debugging and fault location.

1.2 Request Message Body

The request message should be in JSON format (UTF-8 encoding) or empty string. JSON object arrays or other formats are not allowed. The business request parameters should be placed in the request message body.

Request message example:

```
POST /api/robot/controller/tasks HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8
Content-Length: 512
Date: Fri, 26 Mar 2021 06:46:14 GMT

{"warehouseId":"371****b108",……,"zoneCode":"c7a6****371b"}
```

1.3 Encrypted Signature

In the application registration process for third-party applications, if encryption is enabled, a signature should be included in the request. The appSecret mentioned below is the private key used in the application registration. A complete request sample is shown above, where only the request line, specific header fields, an empty line, and the message body are involved in the signature. That is, header fields not involved in the signature should be removed from the original request message, the header fields involved in the signature should be sorted in the sequence shown in the table below, and the header field names should be converted to uppercase.

Header fields involved in the signature:

No.	Header Field	Req. or Opt.
1	AUTHOTIZATION	REQ.
2	HOST	REQ.
3	X-LR-APPKEY	REQ.
4	X-LR-REQUEST-ID	REQ.
5	X-LR-SOURCE	OPT.
6	X-LR-TRACE-ID	OPT.
7	X-LR-VERSION	REQ.

The request sample above is the one after processing, and the original request sample is as follows:

```
POST /api/robot/controller/tasks HTTP/1.1
AUTHORIZATION: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
HOST: 10.10.10.10:1010
X-LR-APPKEY: 75ddbd3e78e64a91a3e68dc7b79ec485
X-LR-REQUEST-ID: d8cdc42a82a3470bb3af766c017703ba
X-LR-SOURCE: wms
X-LR-TRACE-ID: fb09af3e14cc42d48eba1457590da6ac
X-LR-VERSION: v1.0

{"warehouseId":" b1d5fc3663f448ea8be4067dd57a0134"}
```

The generation of signature:

```
Signature = MD5 (salted hash algorithm (appSecret, the concatenated original string of request
```

The following salted hash algorithms should be used:

- HMAC-SHA256 (Recommended)
- HMAC-SHA512

Use the MD5 hash algorithm to hash the salted hash value again to generate a 128-bit (16-byte) hash value.

For example, if *appSecret* is c000aada00554a47aeb988eb05af3153, the generated hash value will be 54fe052cbd443c4561ecab26df8c02c10ce3624b815f5c48b532cfa01fb178cf and after MD5 processing, the signature will be d62f992a5ad0a126.

The request signature should be at the end of the search parameters, with the parameter name being sign, as shown below:

```
POST /api/robot/controller/tasks?sign= d62f992a5ad0a126 HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: 75ddbd3e78e64a91a3e68dc7b79ec485
X-lr-version: v1.0
X-lr-trace-id: fb09af3e14cc42d48eba1457590da6ac
X-lr-request-id: d8cdc42a82a3470bb3af766c017703ba
X-lr-source: wms
Content-Type: application/json;charset=UTF-8
Content-Length: 51

{"warehouseId":" b1d5fc3663f448ea8be4067dd57a0134"}
```

The parameters in the header field Authorization are detailed below.

Parameter	Data Type	Length	Req. or Opt.	Description
nonce	String	8	Req.	A random number. It is recommended to be different for each request to effectively resist rainbow-table based attacks. It can also be changed periodically.
method	String	N/A	Req.	Salted hash algorithm. Fixed enumeration values: HMAC-SHA256 HMAC-SHA512
timestamp	String	N/A	Req.	Request time. The time should be in the format with second precision defined in this document.

During the signature verification, the server should first extract the signature from the search parameters, then generate another signature using the same steps as in the signing process, and compare whether the two are the same.

During the replay attack verification, the server should use the timestamp in the Authorization header field, plus the allowed valid request time for the business, to determine the valid timeout duration for the request, and then compare whether the valid timeout duration has exceeded the current server time. It is recommended that the valid request time set by the server should not exceed 120 seconds.

1.4 Response Header Fields

Security and technical information in the response should be placed in the response header fields to avoid coupling with business parameters.

Default response header fields are detailed below.

Field Name	Data Type	Max. Length	Req. or Opt.	Description
Content-Type	String	N/A	Opt.	Fixed value: application/json;charset=UTF-8
X-lr-request-id	String	16	Req.	Unique request ID.
X-lr-version	String	12	Opt.	API version. To ensure backward compatibility, API versioning is adopted.
X-lr-trace-id	String	32	Opt.	Full-stack tracing ID. It is used to identify upstream and downstream faults, and it should be returned in the response result. UUID is recommended.

1.5 Response Message Body

When the HTTP response status code is 200, the JSON object format of the response message body is as follows:

Name	Field Name	Data Type	Max. Length	Req. or Opt.	Description
Message Code	code	String	32	Req.	See Response Status Codes for definitions of generic response codes and message codes of APIs.
Prompt Message	message	String	256	Opt.	Exception description.
Business Data	data	JSON Object	10 MB	Opt.	Returned business attribute object.

Sample for normal response message body:

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
X-lr-timestamp: 2021-01-01T00:00:00+08:00
```

{"code":"SUCCESS","message":"succeeded","data": {"robotTaskCode": "abc**13"}}

Sample for business exception response message body:

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
X-lr-timestamp: 2021-01-01T00:00:00+08:00

{"code":"Err_Internal","message":"Unknown internal error","data":null}
```

Sample for request exception response message body:

```
HTTP/1.1 403 Forbidden
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
X-lr-timestamp: 2021-01-01T00:00:00+08:00
```

1.6 Response Status Codes

Status Code	Description
200	On the HTTP protocol level, the processing is successful, but business exceptions may still occur and need to be determined based on the message code in the response message body.
401	Signature authentication failed due to invalid signature, expired signature, or invalid appKey and appSecret. It is necessary to update the request time and re-sign, or the system may need to reissue the appKey and appSecret.
403	Insufficient permissions. For example, if a business system attempts to cancel a task not created by the system, the error code will be returned.
406	Invalid Content-Type in the request.
400	Other exceptions caused by client request errors. These exceptions cannot be solved through retries, and it is required to check if the request is in valid format.

500	Server-side exceptions. These exceptions can be resolved by resending the request. The number of retries and intervals should be set based on the actual conditions.

All other unlisted status codes should be handled according to the standard HTTP protocol.

1.7 General Codes for Response Message Body

Code	Message
SUCCESS	Succeeded.
Err_Internal	Unknown internal error.
Err_DataValidationFailed	Validating data format failed.
Err_RequestDuplicate	Request is repeated.
Err_InvalidVersion	Invalid request version.

1.8 Update History

Summary of Changes in Version 4.3_March, 2026

Version	Summary of Changes
Version 4.3_Mar., 2026	1. Edited the API information in ***Task Group***: Changed the type of the request parameter **groupCode** from Opt. to Req.
	2. Edited the API information in ***Apply Task***: Changed the enumeration value of the request parameter **taskType**; Changed the description of the request parameter **deadline**; Added the request parameter **expectedStartTime**; Changed the description of the request parameter **robotCode**; Deleted the response parameter **seq**; Added the response parameter **type**, **code**, **operation**, **robotType**, **robotCode**, and **extra**;

	3. Edited the API information in ***Continue Task***: Deleted the request parameter **robotTaskCode**; Deleted the response parameter **seq**; Added the response parameter **type**, **code**, **operation**, **robotType**, **robotCode**, and **extra**;
	4. Edited the API information in ***Cancel Task***: Changed the description of Task Cancellation API; Added the request parameter **robotCode**; Added the response parameter **extra**.
	5. Edited the API information in ***Set Task Priority***: Changed the description of the request parameter **deadline**; Changed the message code from **Err_TaskNotFound** to **Err_DataValidationFailed**.
	6. Edited the API information in ***Pause and Restore AMRs in Specified Area***: Changed the description of Area AMR Pausing and Restoration API; Changed the description of the request parameter **invoke**.
	7. Edited the API information in ***Return AMR to Fixed Area***: Added the request parameter **mapCode** and **zoneCodes**; Changed the description of the request parameter **bootTime** and **expireTime**.
	8. Edited the API information in ***Clear Area***: Changed the description of Area Clearing API; Changed the type of the request parameter **zoneCode** from Opt. to Req.; Changed the request parameter from **invoke** to **zoneCodes**; Added the request parameter **targetZoneCode**, **report**, and **invoke**;

	Changed the max. length of the request parameter **pause** from None to 2, and changed the description; Changed the message code from **Err_TaskNotFound** to **Err_DataValidationFailed**, from **Err_TargetZoneCodeNotFound** to **ERROR.SYS_INVALID_PARAM**; Deleted the message code **Err_NoTargeMatch** and **Err_NoPathMatch**.
	9. Edited the API information in ***Block and Release Area***: Added the request parameter **pause**, **report**, **disableSite**, and **enableSite**; Changed the message code from **Err_TaskNotFound** to **Err_DataValidationFailed.**
	10. Edited the API information in ***Link Carrier to Point***: Changed the description of Carrier Linkage to Point API.
	11. Edited the API information in ***Unlink Carrier from Point***: Changed the description of Carrier Unlinking from Point API; Changed the type of the request parameter **carrierCode** and **siteCode** from Opt. to Req.
	12. Edited the API information in ***Link or Unlink Storage Object to or from Carrier***: Changed the description of the request parameter **carrierCategory**; Changed the request parameter from **temporary** to **carrierDir**, changed the max. length from 2 to 4, and changed the description; Changed the request parameter from **carrierDir** to **invoke**, changed the data type from integer to string, the max. length from 4 to None, the parameter type from Opt. to Req., and changed the description; Changed the request parameter from **invoke** to **stackLabel**, changed the max. length from None to 32, the data type from Req. to Opt., and changed the description.
	13. Edited the API information in ***Enable or Disable Carrier***: Changed the max. length of request parameter **carrierCode** from 64 to 32;

	Deleted the response parameter extra.
	14. Edited the API information in ***Enable or Disable Point:*** Deleted the response parameter **extra**.
	15. Edited the request parameters in ***Notify of Third-Party Device Execution (Return Code: 0)***: Changed the description of request parameter **eqptCode**; Deleted the response parameter **taskCode** and **extra**.
	16. Edited the API information in ***Apply Pre-Scheduling Task***: Added the request parameter robotTaskCode; Changed the description of request parameter **taskCount** and **capacityCount**. Changed the max. length of request parameter **capacityCount** from 4 to None Changed the response parameter from **taskCode** to **robotTaskCode**.
	17. Deleted the General API.
	18. Edited the API information in ***Search Task Status***: Changed the description of request parameter **taskType** and **warning**; Deleted the response parameter **seq**, and added the response parameter **type**, **code**, **autoStart**, **operation**, **robotCode**, and **extra**.
	19. Edited the API information in ***Search AMR Status***: Changed the description of response parameter **warnings**.
	20. Edited the API information in ***Search Carrier Status***: Changed the description of Carrier Status Searching API; Changed the type of response parameter **x, y,** and **carrierDir** from Opt. to Req.; Changed the description of response parameter **x** and **y**; Added the response parameter **binCode**, **binName**, and **robotCode**;

	21. Added the Carrier Linkage to Materials API, Carrier Unlinking from Materials, and Third-Party Execution Notification API in ***Link Carrier to Material***, ***Unlink Carrier from Material***, ***and*** ***Notify of Third Party Execution (Return Code: SUCCESS)***.
	22. Edited the API information in ***Receive Feedback on Task Performance***: Added the response parameter **amrCode**, **layerNo**. And **carrierWeight**.
	23. Deleted the Traffic Control Area Application and Releasing API.
	24. Edited the API information in ***Request for Resource***: Changed the description of request parameter **applyType**, **resourceType**, and **label**; Changed the type of response parameter **data** from Opt. to Req.
	25. Edited the API information in ***Request for Third-Party Device***: Changed the description of request parameter **eqptCode**.
	26. Edited the API information in ***Receive Feedback on Completion of AMR Returning***: Changed the description of AMR Returning Feedback API; Deleted the request parameter **mapCode**, **shutDownTime**, and **timeToLive**; Changed the request parameter from **mode** to **workRobotCode**, from **homingZone** to **status**; Changed type of request parameter **workRobotCode** from sting to array of strings, the data type from Req. to Opt.; Changed the data type of request parameter **status**. Deleted the response parameters.
	27. Edited the API information in ***Receive Feedback on Completion of Area Clearing***: Changed the description of Area Clearing Feedback API; Deleted the request parameter **mapCode**, **invoke**, **report**, **pause**, and **controlMode**;

	Changed the request parameter from **data** to **areaList** and from **targetZoneCode** to **stayRobotCode**; Changed the description of request parameter **status**; Deleted the object type data, **targetZoneCode**, **posUpLeft**, and **posDownRight**; Added the object type **areaList** and the response parameter **areaCode**.
	28. Edited the API information in ***Report AMR Exception Alarm***: Changed the description of AMR Exception Alarm Reporting API.
	29. Edited the API information in ***Report Task Exception Alarm***: Changed the description of Task Exception Alarm Reporting API; Added the request parameter **mapCode**.
	30. Added the Third-Party Linking/Unlinking Notification API.
	31. Added the FlashStation Tote Inbound Verification API.
	32. Added scene 6 and scene 7 in ***Typical Scheduling Scenes***: Added Scene 6: Call a LMR to perform an outbound in sequence task; Added Scene 6: Call a LMR to perform an outbound in sequence task.

Summary of Changes in Version 4.2_January, 2025

New document for RCS-2000 V4.2.

Chapter 2 API Reference

2.1 AMR Scheduling APIs

2.1.1 Task Group

API Name	Task Group API				
Description	1. A task group is a collection of tasks. When there are interrelationships between tasks, it is necessary to first call the task group API to describe the strategy and the combination of relations between tasks, and then call the task application API. 2. Task strategies include outbound by group sequence and allocation by groups. 3. The rule for applying sequential outbound tasks: The upper level system calls the API to apply the *groupSeq* from small to large to RCS-2000.				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/task/group				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	groupCode	String	32	Req.	Task group No., which is globally unique.

strategy	String	16	Req.	Execution strategy. The enumeration can be extended. Predefined enumeration values: GROUP_SEQ: outbound by groups in sequence, GROUP_ASSIGN: task assignment by groups (for CTU workstation tasks only), GROUP_CARRIER_ADJUST: carrier tallying (for CTU only, used to tally carriers to appropriate bins).
strategyValue	String	32	Opt.	The following strategies are required when the value of *strategy* is GROUP_SEQ. Types of sequential outbound: 0: unordered between and within groups and outbound by groups, 1: ordered within and between groups, 2: ordered between groups and unordered within groups, 3: unordered between groups and ordered within groups. GROUP_CARRIER_ADJUST The following strategy is required for CTU tallying. 3: unordered between groups and ordered within groups.
groupSeq	Integer	8	Opt.	Group sequence (number), from 1 to 9999999999.
targetRoute	JSON Object	N/A	Opt.	The next target position for task execution. It is used to control the sequence of tasks assigned to one workstation. If it is not provided, only global sequence

					can be used for controlling task sequence to one workstation
	data	Array of JSON Objects	N/A	Req.	
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	code	String	N/A	Req.	Message code.
	message	String	N/A	Req.	Message content.
Dedicated Message Code	code		Description		
	Err_TaskNotStart		Last group of tasks has not been applied.		
	Err_DataValidationFailed		Incorrect task sequence No.		
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
targetRoute	type	String	16	Req.	SITE: point alias, Zone: target area No.
	code	String	256	Req.	
data	robotTaskCode	String	64	Req.	Task No., which is globally unique.
	sequence	Digits	N/A	Opt.	Task sequence (number), from 1 to 9999. It is not needed if task sequence is not required within a task group.

Request Sample

```
POST /rcs/rtas/api/robot/controller/task/group HTTPS/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "groupCode": "2e0d1ae0481f48b78b6a217ac2b54eb4",
    "strategy": "GROUP_SEQ",
    "strategyValue": "1",
    "groupSeq": 10,
     "targetRoute": {
        "type": "ZONE",
        "code": "A2"
},
    "data": [
        {
            "robotTaskCode": "0a17e361eb5248bfab0508e4709f085e",
            "sequence": 1
        },
        {
            "robotTaskCode": "1b30143f32914155ab194d55a95d54da",
            "sequence": 2
        },
        {
            "robotTaskCode": "6541e925d1de4c448188068fcee75de5",
            "sequence": 3
        }
    ]
}
```

Response Sample

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
}
```

2.1.2 Apply Task

API Name	Task Applying API				
Description	It is called by business system to send task request, and then RCS-2000 will generate task execution orders for execution.				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/task/submit				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description

	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	taskType	String	16	Req.	Task type. It is configured in RCS-2000, and it encapsulates internal scheduling logic that is independent of the business system. The enumeration can be extended. Predefined enumeration value: PF-LMR-COMMON: carrying, containing one or several execution steps.
	targetRoute	Array of objects	N/A	Req.	Collection of execution steps. The key route of the AMR to execute the task, for example, rack start point A to outbound workstation point B.
	initPriority	Integer	8	Opt.	Initial priority of the task, with larger value representing higher priority. The system will make a comprehensive assessment and dynamically adjust the actual priority based on the task busyness level, AMR working status, and other priority adjustment conditions. Range: 1 to 120.
	deadline	Time	N/A	Opt.	Task deadline, in second. It is one of the task priority adjustment conditions. Task should be completed before the deadline. The system will prompt for the task timeout. Sample: 2021-04-04T12:23:55Z
	expectedStartTime	Time	N/A	Opt.	Expected task start time. This field is used when applying a task to insert the task into a previous time point for execution, meaning subsequent task assignments need to prioritize

				this task. Sample: 2021-04-04T12:23:55Z. The longer the interval from the current time, the higher the priority. The expected start time cannot be earlier than the task deadline.
robotType	String	N/A	Opt.	The AMR is selected from the specified range to perform the task. The business system ensures the correct AMR type used in the task. If AMR selection range is not specified manually, the scheduling system will look for the optimal solution within all available AMRs. Fixed enumeration values: GROUPS: AMR resource group No. ROBOTS: AMR No.
robotCode	Array of strings	N/A	Opt.	The unique ID of the resource type that matches with the *robotType*. Use comma (,) to separate multiple numbers. Each AMR No.: 1 to 64 characters.
interrupt	Integer	1	Opt.	Whether the task can be interrupted. 1: interruption allowed. If the carrier is assigned with other tasks, the current task can be interrupted. 0: interruption not allowed. The current task cannot be interrupted even if the carrier is assigned with other tasks.
robotTaskCode	String	64	Opt.	The unique No. of the external task. If the value is empty, the system will generate a task No. and return it.
groupCode	String	32	Opt.	Task group No., which is globally unique.
extra	JSON Object	N/A	Opt.	Custom extension field used to pass different extended contents according to business differences. It

					can be used during the execution of business processes or for transparent transmission.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	code	String	N/A	Req.	Message code.
	message	String	N/A	Req.	Message content.
	data robotTaskCode	String	64	Req.	Task No., which is globally unique.
	data extra	JSON Object	N/A	Opt.	Custom extension field (reserved).
Dedicated Message Code	code	Description			
	Err_TaskTypeNotSupport	Task type is not supported.			
	Err_RobotGroupsNotMatch	Scheduling failed. AMR resource group No. does not match with the task.			
	Err_RobotCodeNotMatch	Scheduling failed. AMR No. does not match with the task.			
	Err_TargetRouteError	Incorrect task route parameters.			
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	type	String	16	Req.	Target type. The enumeration can be extended.

				Predefined enumeration values: ZONE: target area No., SITE: point alias, STORAGE: bin alias, MAT_LABEL: material label, CARRIER: carrier No., STACK: roadway No., STACK_LABEL: roadway attribute value, CHANNEL: aisle No., EQPT: third-party device, PTL_WALL: buffer rack, CARRIER_TYPE: carrier type, BIN_TYPE: bin type, AREA_STATION: area workstation, PILE_COUNT: carrying quantity at one time, MIX_CONDITION: mixed conditions, DEGREE: distance from point to workstation, MULTI_BIN_GROUP: multi deep bin group No.
code	String (JSON array)	256	Req.	The target No. corresponding to *type*. For ZONE enumeration, it supports enumerating multiple areas separated by commas (,), for example, ZONE1, ZONE2. For MIX_CONDITION enumeration, the sample codes are as follows: [{"type":"ZONE","code":"SF2"},{"type":"CARRIER_TYPE","code":"P1"}]
operation	String	8	Opt.	Operation after AMR reaches the target position. It is optional and has no effect on AMR actions. The enumeration can be extended.

					Predefined enumeration values: COLLECT: picking, DELIVERY: putaway, ROTATE: rotation.
	robotType	String	N/A	Opt.	The AMR is selected from the specified range to perform the task. The business system ensures the correct AMR type used in the task. If the AMR selection range is specified manually for the current task, the scheduling system will ignore the range set in the task; if the AMR selection range is neither specified manually nor specified in the task, the scheduling system will look for the optimal solution within all available AMRs. If not referenced in the process, this value takes no effect. Fixed enumeration values: GROUPS: AMR resource group No., ROBOTS: AMR No.
	robotCode	Array of strings	N/A	Opt.	The unique ID of the resource type that matches with the *robotType*.
	extra	JSON Object	N/A	Opt.	Custom extension field to pass different extended contents according to business differences. It can be used during the execution of business processes.

JSON Object Structure

Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
extra	angleInfo	JSON Object	N/A	Opt.	Angle information, including angle type and angle value.

	carrierInfo	Array of JSON Objects	N/A	Opt.	Carrier information, including carrier No. and layer No. Application scenes: ● specially for loader/unloader or conveyor; ● where the information of carrier that is not linked to a point is required.
	crossVisionDoor	String	2	Opt.	Whether the AMR passes through the vision gate. If not provided, AMR defaults to "not pass." Set this value to 1 for AMR to pass through the gate without stopping. This parameter is valid only when placed in the top-level extra object. 0: not passed, 1: passed.
	pickStationCode	String	16	Opt.	Picking station No. It identifies the picking station for each outbound task. This parameter is valid only when placed in the top-level extra object.

JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
angleInfo	type	String	32	Opt.	Angle type. The enumeration can be extended. Predefined enumeration value: ABSOLUTE: absolute angle, which is dependent on the workstation direction.

	code	String	64	Opt.	The value corresponding to *type*. Angle: 0, 90, 180, -90, 360.
carrierInfo	carrierType	String	16	Opt.	Carrier type.
	carrierCode	String	16	Opt.	Carrier No.
	layer	Integer	N/A	Opt.	Layer No., bottom-up numbering starting from 0.

Request Sample

```
P POST /rcs/rtas/api/robot/controller/task/submit HTTPS/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "taskType": "PF-LMR-COMMON",
    "targetRoute": [
        {
            "type": "ZONE",
            "code": "7256****7a12",
            "operation": "COLLECT",
            "extra": null
            "robotType": "GROUPS",
            "robotCode": ["a361****f1b8"]
        },{
```

```
                "type": "SITE",
                "code": "c9a1****3a1c",
                "operation": "DELIVERY",
                "extra": {
                    "angleInfo": {
                        "type": "RELATIVE",
                        "code": "90",
                    },
                    "carrierInfo": [{
                        "carrierType": "1",
                        "carrierCode": "7256****7a12",
                        "layer": "0"
                    }]
                }
            }
        ],
        "initPriority": 99,
        "deadline": "2021-04-04T12:23:55Z",
        "robotType": "GROUPS",
        "robotCode": ["c3a1****e7bb"],
        "extra": {
                    "carrierInfo": [{
                        "carrierType": "1",
                        "carrierCode": "7256****7a12",
                        "layer": "0"
                    }]
            }
}
```

Response Sample

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "robotTaskCode": "27e****ba08",
        "extra": null
    }
}
```

2.1.3 Continue Task

API Name	Task Continuation API
Description	One task contains several steps. This API is called to start the first step, and after each step is completed, it should be called by the business system to drive the next step. If the step is set to start automatically when the task is applied, the API is not required, but if the scheduling system receives the command to continue executing the step, the command should be handled according to the idempotent principle For steps being executed, repeated calls should be handled according to the idempotent principle. For command on steps that have been completed or skipped, an error should be reported.
Request	
Provider	RCS-2000
Service Prefix	/rcs/rtas
Path	/api/robot/controller/task/extend/continue

Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	triggerType	String	32	Req.	Triggering type: SITE: triggering by point No., CARRIER: triggering by carrier No., ROBOT: triggering by AMR No., TASK: triggering by task chain No.
	triggerCode	String	64	Req.	The triggering No. corresponding to *triggerType*.
	targetRoute	Object	N/A	Opt.	The next target position for task execution.
	extra	Object	N/A	Opt.	Custom extension parameters.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message	Field Name	Data Type	Max. Length	Req. or Opt.	Description

Body Business Data	code	String	N/A	Req.	Message code.
	message	String	N/A	Req.	Message content.
Dedicated Message Code	code		Description		
	Err_TaskNotStart		Task not started.		
	Err_TaskFinished		Task ended.		
	Err_TaskNotFound		Task not found.		
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	type	String	16	Req.	Target type. The enumeration can be extended. Predefined enumeration values: ZONE: target area No., SITE: point alias, STORAGE: bin alias, MAT_LABEL: material label, CARRIER: carrier No., STACK: roadway No., STACK_LABEL: roadway attribute value, EQPT: third-party device, PTL_WALL: buffer rack, CARRIER_TYPE: carrier type, BIN_TYPE: bin type, AREA_STATION: area workstation, PILE_LAYER: stacking layer No., PILE_COUNT: carrying quantity at one time, MIX_CONDITION: mixed conditions.

code	String (JSON array)	1000	Req.	The target No. corresponding to *type*. For ZONE enumeration, it supports enumerating multiple areas separated by commas (,), for example, ZONE1, ZONE2. For MIX_CONDITION enumeration, the sample codes are as follows: [{"type":"ZONE","code":"SF2"},{"type":" CARRIER_TYPE","code":"P1"}]
operation	String	8	Opt.	Operation after AMR reaches the target position. The enumeration can be extended. Predefined enumeration values: COLLECT: picking, DELIVERY: putaway, ROTATE: rotation.
robotType	String	N/A	Opt.	The AMR is selected from the specified range to perform the task. The business system ensures the correct AMR type used in the task. If the AMR selection range is specified manually for the current task, the scheduling system will ignore the range set in the task; if the AMR selection range is neither specified manually nor specified in the task, the scheduling system will look for the optimal solution within all available AMRs. Fixed enumeration values: GROUPS: AMR resource group No., ROBOTS: AMR No.

	robotCode	Array of strings	N/A	Opt.	The unique ID of the resource type that matches with the *robotType*.
	extra	JSON Object	N/A	Opt.	Custom extension field of *targetRoute* to pass different extended contents according to business differences. It can be used during the execution of business processes or for transparent transmission.
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
extra	angleInfo	JSON Object	N/A	Opt.	Angle information, including angle type and angle value.
	carrierInfo	Array of JSON Objects	N/A	Opt.	Carrier information, including carrier No. and layer No. Application scenes: 1) specially for loader/unloader or conveyor; 2) where the information of carrier that is not linked to a point is required.
	crossVisionDoor	String	2	Opt.	Whether the AMR passes through the vision gate. If not provided, AMR defaults to "not pass." Set this value to 1 for AMR to pass through the gate without stopping. This parameter is valid only when placed in the top-level extra object. 0: not pass 1: pass

	pickStationCode	String	16	Opt.	Picking Station No. . Identifies the picking station for this outbound task. This parameter is valid only when placed in the top-level extra object.
JSON Object Structure					
angleInfo	type	String	32	Opt.	Angle type. The enumeration can be extended. Predefined enumeration value: ABSOLUTE: absolute angle, which is dependent on the workstation direction.
	code	String	64	Opt.	The value corresponding to *type*. Angle value: [0, 90, 180, -90, 360].
carrierInfo	carrierType	String	16	Opt.	Carrier type.
	carrierCode	String	16	Opt.	Carrier No.
	layer	Integer	N/A	Opt.	Layer No., bottom-up numbering starting from 0.

Request Sample

```
POST    /rcs/rtas/api/robot/controller/task/extend/continue HTTPS/1.1

Host: 10.10.10.10:1010

Authorization:          nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"

X-lr-appkey: a4f********b324

X-lr-version: v1.0
```

```
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "triggerType":"TASK",
     "triggerCode": "1a17e361eb5248bfab0508e4709f085f5",
     "targetRoute": {
          "type":"ZONE",
          "code":"D1",
          "operation": "COLLECT"
    }
 }
```

Response Sample

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "robotTaskCode": "27e****ba08",
        "nextSeq": 1,
        "extra": null
    }
}
```

2.1.4 Cancel Task

API Name	Task Cancelation API				
Description	This API can be called by the business system to send task cancelation request, and RCS-2000 will cancel the current task and return the result. When canceling the current task, you can also apply a new task to the AMR via this API. If the canceled task is still in the pending assignment status, the new task will also be created, and the priority and AMR selection range of the canceled task will be inherited. The task No. can be specified in *extra* parameter; otherwise, the RCS-2000 will generate one randomly. The task cancellation API now supports batch cancellation of tasks by *robotCode* or *carrierCode*: 1. When a Task No. is provided, the task will be canceled and processed according to the original logic; 2. An error occurs if the Task No., carrier No., and AMR No. are all empty; 3. When tasks are canceled in bulk by AMR No. or carrier No., the cancellation type must be DROP; 4. When tasks are canceled only by AMR No., CTU No. must be excluded; 5. Cancellation is not permitted if the carrier No. or AMR No. of a vehicle incomplete subtask in the task chain does not match the input parameters.				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/task/cancel				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				

	Parameter	Data Type	Max. Length	Req. or Opt.	Description
Request Message Body	robotTaskCode	String	64	Opt.	Task No., which is globally unique.
	cancelType	String	16	Req.	Task cancelation type. CANCEL: cancelation (soft cancelation), DROP: manual intervention (hard cancelation).
	carrierCode	String	16	Opt.	Returning carrier No.
	robotCode	String	16	Opt.	AMR No.
	reason	String	128	Opt.	Reason for cancelation.
	returnTaskType	String	N/A	Opt.	Returning task type after soft cancelation. The task type is required for soft cancelation. The PF-TASK-CANCEL-RETURN process is initialized for LMRs, FMRs and CTUs in RCS-2000. For special requirements, you can configure other returning processes after soft cancelation.
	targetRoute	JSON Object	N/A	Opt.	
	extra	JSON Object	N/A	Opt.	Custom extension field.

Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				

	Field Name		Data Type	Max. Length	Req. or Opt.	Description
Response Message Body Business Data	code		String	N/A	Req.	Message code.
	message		String	N/A	Req.	Message content.
	data	robotTaskCode	String	64	Req.	The unique ID of the canceled task.
		extra	JSON Object	N/A	Opt.	Custom extension field.

	code	Description
Dedicated Message Code	Err_TaskFinished	Task ended.
	Err_TaskNotFound	Task not found.
	Err_TaskModifyReject	Task cannot be changed.
	Err_TaskTypeNotSupport	New task type not supported.
	Err_RobotGroupsNotMatch	Scheduling failed. AMR resource group No. does not match with the new task.
	Err_RobotCodesNotMatch	Scheduling failed. AMR No. does not match with the new task.

JSON Object Structure

Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
targetRoute	type	String	16	Req.	Target type. The enumeration can be extended. Predefined enumeration values: ZONE: target area No., SITE: point alias, STORAGE: bin alias, MAT_LABEL: material label, CARRIER: carrier No., STACK: roadway No., STACK_LABEL: roadway attribute value,

					EQPT: third-party device, PTL_WALL: PTL, CARRIER_TYPE: carrier type, BIN_TYPE: bin type, AREA_STATION: area workstation, PILE_LAYER: stacking layer No., PILE_COUNT: carrying quantity at one time, MIX_CONDITION: mixed conditions.
	code	String (JSON array)	1000	Req.	The target No. corresponding to *type*. For ZONE enumeration, it supports enumerating multiple areas separated by commas (,), for example, ZONE1, ZONE2. For MIX_CONDITION enumeration, the sample codes are as follows: [{"type":"ZONE","code":"SF2"},{" type":" CARRIER_TYPE","code":"P1"}]
extra	taskCode	String	64	Opt.	Task chain No. of the returning task generated by soft cancelation

Request Sample

```
POST /rcs/rtas/api/robot/controller/task/cancel HTTPS/1.1

Host: 10.10.10.10:1010

Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"

X-lr-appkey: a4f********b324
```

```
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "robotTaskCode": "27e****ba08",
     "cancelType": "CANCEL",
    "reason": "more urgent tasks",
    "returnTaskType": "PF-TASK-CANCEL-RETURN",
    "targetRoute": {
        "type": "ZONE",
        "code": "7256****7a12"
    },
    "extra": {
         "taskCode": "76****12"
     }
}
```

Response Sample

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
```

```
    "data": {
        "robotTaskCode": "27e****ba08",
        "extra": null
    }
}
```

2.1.5 Set Task Priority

API Name	Task priority setting API				
Description	After the task is created and before the task ends, the API can be called to set parameters related to the task priority, including the initial priority and deadline. RCS-2000 will make a comprehensive assessment and dynamically adjust the actual priority based on the task busyness level, AMR working status, and other priority adjustment conditions.				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/task/priority				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				

	Parameter	Data Type	Max. Length	Req. or Opt.	Description
Request Message Body	robotTaskCode	String	64	Req.	Task No., which is globally unique.
	initPriority	Integer	8	Req.	Initial priority of the task, with larger value representing higher priority. The system will make a comprehensive assessment and dynamically adjust the actual priority based on the task busyness level, AMR working status, and other priority adjustment conditions. Range: 1 to 120.
	deadline	Time	N/A	Opt.	Task deadline, unit: sec. It is one of the task priority adjustment conditions. Task should be completed before the deadline. Sample: 2022-06-14T18:48:14.4643807
	extra	JSON Object	N/A	Opt.	Custom extension field.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message	Field Name	Data Type	Max. Length	Req. or Opt.	Description

Body Business Data	code		String	N/A	Req.	Message code.
	message		String	N/A	Req.	Message content.
	data	robotTaskCode	String	64	Req.	Task No., which is globally unique.
		extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description			
	Err_TaskFinished		Task ended.			
	Err_DataValidationFailed		Task not found.			
JSON Object Structure						
Object Type	Parameter		Data Type	Max. Length	Req. or Opt.	Description
N/A						

Request Sample

```
POST /rcs/rtas/api/robot/controller/task/priority HTTPS/1.1

Host: 10.10.10.10:1010

Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"

X-lr-appkey: a4f********b324

X-lr-version: v1.0

X-lr-trace-id: 605****8a0b

X-lr-request-id: 393****a6c1

Content-Type: application/json;charset=UTF-8


{
```

```
    "robotTaskCode": "27e****ba08",
    "initPriority": 19,
    "deadline": "2021-04-05T12:23:55Z",
    "extra": null
}
```

Response Sample

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "robotTaskCode": "27e****ba08"
        "extra": null
    }
}
```

2.1.6 Pause and Restore AMRs in Specified Area

API Name	Area AMR Pausing and Restoration API
Description	The API can be called to pause or restore AMRs within the designated area.
Request	
Provider	RCS-2000

Service Prefix	/rcs/rtas				
Path	/api/robot/controller/zone/pause				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	zoneCode	String	32	Req.	Specified scheduling control area No., which is globally unique.
	mapCode	String	16	Opt.	Map No. If the area is a temporary area, map No. is required. This parameter is only used for MonitorClient.
	invoke	String	N/A	Req.	Fixed enumeration values: FREEZE: stop, RUN: restore. CHASSIS_PAUSE: pause the chassis only (used for CMR only).

Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	code	String	N/A	Req.	Message code.
	message	String	N/A	Req.	Message content.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	Err_StatusNotMatch		Converting the AMR status to target status failed due to an exception.		
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description

Request Sample

```
POST /rcs/rtas/api/robot/controller/zone/pause HTTPS/1.1

Host: 10.10.10.10:1010

Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"

X-lr-appkey: a4f********b324
```

```
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "zoneCode": "27e****ba08",
     "invoke": "FREEZE",
    "extra": null
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "extra": null
    }
}
```

2.1.7 Return AMR to Fixed Area

API Name	Returning AMR to Fixed Area API				
Description	This API can be called to return the AMRs in the specified area to predefined parking area. It supports setting AMRs to shut down immediately when they reaches the parking position and to power on automatically at the set time during the long-term vacation of the plant.				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/zone/homing				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	mapCode	String	16	Opt.	Map No. Map No., area No., and area No. collection cannot be empty simultaneously. If only map No. is provided without

					area No., all AMRs on the map will return to the fixed area.
	zoneCode	String	32	Opt.	Specified scheduling control area No., which is globally unique. The AMRs in the area will return to the fixed area.
	zoneCodes	Array of strings	N/A	Opt.	The area No. collection of the specified scheduling control areas which must belong to the same map.
	autoShutdown	String	N/A	Req.	Whether AMRs shut down after reaching the fixed area. Fixed enumeration values: YES: shut down, NO: not shut down.
	bootTime	Time	N/A	Opt.	Preset startup time, unit: min. It takes effect when the value of *autoShutdown* is YES.
	expireTime	Time	N/A	Opt.	Timeout duration for returning to fixed area, unit: sec (10 minutes by default).
	extra	JSON Object	N/A	Opt.	Custom extension field.

Response	
Status Code	Description
200	The request succeeded.

Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	homingCode	String	32	Req.	Returning command No.
	robotCount	Integer	32	Req.	The number of AMRs receiving the returning command.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	Err_TimeMachineRequired		The preset startup time is earlier than the current time.		
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /rcs/rtas/api/robot/controller/zone/homing HTTPS/1.1

Host: 10.10.10.10:1010

Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"

X-lr-appkey: a4f********b324

X-lr-version: v1.0

X-lr-trace-id: 605****8a0b
```

```
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "zoneCode": "821****8ac7",
    "autoShutdown": "YES",
    "bootTime": "2021-04-06T08:00:00+8:00",
    "expireTime": "2021-04-05T12:30:00Z",
    "extra": null
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "homingCode": "545****8b0e",
        "robotCount": 20,
        "extra": null,
    }
```

```
}
```

2.1.8 Clear Area

API Name	Area Clearing API				
Description	This API can be called to move AMRs, including AMRs performing tasks, out of the specified area and prohibit other AMRs to enter the area.				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/zone/banish				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	zoneCode	String	32	Opt.	The specified scheduling control area No., which is globally unique. AMRs in the area will move out of the area.
	zoneCodes	Array of Strings	N/A	Opt.	The area collection and area No. cannot be both empty. The collection of areas must be in the same map.

targetZoneCode	String	32	Opt.	The target area where the AMR is moving. It can be selected freely if empty.
pause	String	2	Opt.	Whether to pause all AMRs out of the area after the area is cleared. 0: not pause (default), 1: pause.
report	String	2	Opt.	Whether to notify the upper level system after completion. 0: not notify (default), 1: notify.
controlMode	String	2	Opt.	Scheduling mode: 0: schedule AMR out of area (default), 1: schedule AMR to temporary parking area out of area, 2: schedule AMR to specified area. Specified area is required.
expireTime	Time	N/A	Opt.	Timeout duration for AMR clearing, unit: sec.
invoke	String	32	Opt.	Area status. If the field does not appear, the default value is BANISH, otherwise, the field is required. Fixed enumeration values: BANISH: clear area, RUN: restore.
extra	JSON Object	N/A	Opt.	Custom extension field.

Response	
Status Code	Description
200	The request succeeded.

Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	code	String	N/A	Req.	Message code.
	message	String	N/A	Req.	Message content.
	banishCode	String	32	Req.	Area clearing command No.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	Err_DataValidationFailed		Area to be cleared not found.		
	ERROR.SYS_INVALID_PARAM		Target area not found.		
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /rcs/rtas/api/robot/controller/zone/banish HTTPS/1.1

Host: 10.10.10.10:1010

Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"

X-lr-appkey: a4f********b324

X-lr-version: v1.0

X-lr-trace-id: 605****8a0b

X-lr-request-id: 393****a6c1

Content-Type: application/json;charset=UTF-8
```

```
{
    "zoneCode": "27e****ba08",
    "targetZoneCode": "27e****ba09",
    "expireTime": "2021-04-05T12:30:00Z",
    "invoke:"BANINSH",
    "extra": null
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "banishCode": "27e****ba08"
        "extra": null
    }
}
```

2.1.9 Block and Release Area

API Name	Area Blocking and Releasing API
Description	This API is only valid for AMRs outside the specified area to prevent them from entering the area, and AMRs in the specified area are not required to leave the area.
Request	
Provider	RCS-2000

Service Prefix	/rcs/rtas				
Path	/api/robot/controller/zone/blockade				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	zoneCode	String	32	Req.	No. of the scheduling control area to be blocked, which is globally unique.
	mapCode	String	16	Opt.	Map No. If the area is a temporary area, map No. is required. This parameter is only used for MonitorClient.
	invoke	String	N/A	Req.	Area status. Fixed enumeration values: BLOCKADE: block, OPENUP: release.
	pause	String	2	Opt.	Whether to pause all AMRs in the area after area blocking is completed. 0: not pause (default), 1: pause, 2: pause chassis only (used for CMR only).

	report	String	2	Opt.	Whether to report to the business system after area blocking is completed. 0: not report (default), 1: report.
	disableSite	Number	N/A	Opt.	Whether to disable point. 0: no, 1: yes, It is valid when *invoke* is set to BLOCKADE. The default value is 1.
	enableSite	Number	N/A	Opt.	Whether to enable point. 0: no, 1: yes, It is valid when *invoke* is set to OPENUP. The default value is 1.

Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	code	String	N/A	Req.	Message code.
	message	String	N/A	Req.	Message content.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	Err_DataValidationFailed		Area to be blocked not found.		
JSON Object Structure					

Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /rcs/rtas/api/robot/controller/zone/blockade HTTPS/1.1

Host: 10.10.10.10:1010

Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"

X-lr-appkey: a4f********b324

X-lr-version: v1.0

X-lr-trace-id: 605****8a0b

X-lr-request-id: 393****a6c1

Content-Type: application/json;charset=UTF-8


{
    "zoneCode": "27e****ba08",
    "invoke": "BLOCKADE"
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
```

```
    "data": {
        "banishCode": "27e****ba08"
        "extra": null
    }
}
```

2.1.10 Link Carrier to Point

API Name	Carrier Linkage to Point API				
Description	This API can be called to link carriers to points. The linkage indicates that the carrier is placed on the point, and the precondition is that both the carrier and point are not occupied by tasks (This API can only be used after the carrier is added in the scheduling system).				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/carrier/bind				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	carrierCode	String	64	Req.	Carrier No. or carrier alias.
	siteCode	String	64	Req.	Point No. or point alias.
	carrierDir	Integer	4	Opt.	Carrier direction, range: [0, 90, 180, -90, 360]. For rack direction, the horizontal and

					vertical directions should be the same as those of the point. If the point is omnidirectional, the direction should be specified in the API. It is not required for square racks.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	code	String	N/A	Req.	Message code.
	message	String	N/A	Req.	Message content.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	Err_TaskFound		Linking or unlinking failed. Carrier or point has a task to be executed.		
	Err_Bound		Carrier or point has been linked to another object.		
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /rcs/rtas/api/robot/controller/carrier/bind HTTPS/1.1
Host: 10.10.10.10:1010
```

```
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "carrierCode": "f51c****ba08",
    "siteCode": "ba46****ec31",
    "extra": null
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "extra": null
    }
}
```

2.1.11 Unlink Carrier from Point

API Name	Carrier Unlinking from Point API
Description	This API can be called to unlink a carrier from a point. Unlinking indicates the carrier is no more placed on the point. At least one of the carrier No. or point No. is required. If both values are passed, and the actual linked carrier

	No. and the input carrier No. mismatch, an error will be reported. If the point is not linked to the carrier, the API calling will be treated as a repeated unlinking (This API can only be used after the carrier is added in the scheduling system).				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/carrier/unbind				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	carrierCode	String	64	Opt.	Carrier No. or carrier alias.
	siteCode	String	64	Opt.	Point No. or point alias.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				

Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	code	String	N/A	Req.	Message code.
	message	String	N/A	Req.	Message content.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	N/A				
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /rcs/rtas/api/robot/controller/carrier/unbind HTTPS/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
"carrierCode": "f51c****ba08",
"siteCode": "ba46****ec31",
    "extra": null
}
```

Response Sample

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
```

```
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "extra": null
    }
}
```

2.1.12 Link or Unlink Storage Object to or from Carrier

API Name	Linking/Unlinking Storage Object to/from Carrier API				
Description	This API can be called to link or unlink storage objects to or from carriers. The linkage indicates that the carrier is placed on the storage object, and the precondition for linkage is that both the carrier and storage object are not occupied by tasks.				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/site/bind				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				

	Parameter	Data Type	Max. Length	Req. or Opt.	Description
Request Message Body	slotCategory	String	16	Opt.	Storage object category. For unlinking, only carrier No. is required, and for linking, this parameter is required. Predefined enumeration values: SITE: point, BIN: bin.
	slotCode	String	64	Opt.	No. or name corresponding to *slotCategory*. For unlinking, only carrier No. or name is required, and for linking, this parameter is required.
	carrierCategory	String	16	Opt.	Carrier category. Predefined enumeration values: POD: rack, PALLET: pallet, BOX: tote.
	carrierType	String	16	Opt.	Carrier type.
	carrierCode	String	64	Opt.	Carrier No. or carrier name.
	carrierDir	Integer	4	Opt.	Rack direction relative to the point, range: [0, 90, -90, 180, 360]. For rack direction, the horizontal and vertical directions should be the same as those of the point. If the point is omnidirectional, the direction should be specified in the API. It is not required for square racks.
	invoke	String	N/A	Req.	Linking or unlinking. Fixed enumeration values: BIND: link,

					UNBIND: unlink.
	stackLabel	String	32	Opt.	Roadway attribute value. It should be consistent with the roadway attribute value of points. Otherwise, an error will be reported.
	colCount	Integer	4	Opt.	Column No. to be linked. It is valid in stacking scenes.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	code	String	N/A	Req.	Message code.
	message	String	N/A	Req.	Message content.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	Err_TaskFound		Linking or unlinking failed. Carrier has a task to be executed.		
	Err_Bound		Carrier has been linked to another object.		
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /rcs/rtas/api/robot/controller/site/bind HTTPS/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
"slotCategory": "SITE",
"slotCode": "000010AA000100",
"invoke": "UNBIND",
    "extra": null
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "extra": null
    }
}
```

2.1.13 Enable or Disable Carrier

API Name	Carrier Enabling or Disabling API

Description	This API can be called to enable or disable carriers. When a carrier is disabled, the carrier and its linked point will not be assigned with tasks, and the tasks that have been assigned should be completed. When the carrier is enabled, the carrier and its linked point can be assigned with tasks.				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/carrier/lock				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	carrierCode	String	32	Req.	Carrier No.
	invoke	String	N/A	Req.	Carrier enabling or disabling. Fixed enumeration values: LOCK: disable, UNLOCK: enable.

Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	code	String	N/A	Req.	Message code.
	message	String	N/A	Req.	Message content.
Dedicated Message Code	code		Description		
	N/A				
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /rcs/rtas/api/robot/controller/carrier/lock HTTPS/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8
```

```
{
"carrierCode": "f51c****ba08",
"invoke": "LOCK"
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded"
}
```

2.1.14 Enable or Disable Point

API Name	Point Enabling or Disabling API
Description	This API can be called to enable or disable points. When a point is disabled, the point and its linked carrier will not be assigned with tasks, and the tasks that have been assigned should be completed. When the point is enabled, the point and its linked carrier can be assigned with tasks.
Request	
Provider	RCS-2000
Service Prefix	/rcs/rtas
Path	/api/robot/controller/site/lock
Idempotence	Yes

Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	siteCode	String	64	Req.	Point No. or point alias.
	invoke	String	N/A	Req.	Point enabling or disabling. Fixed enumeration values: LOCK: disable, UNLOCK: enable.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	code	String	N/A	Req.	Message code.

Data	message	String	N/A	Req.	Message content.
Dedicated Message Code	code		Description		
	N/A				
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /rcs/rtas/api/robot/controller/site/lock HTTPS/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
"siteCode": "ba46****ec31",
"invoke": "LOCK"
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
```

```
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded"
}
```

2.1.15 Notify of Third-Party Device Execution (Return Code: 0)

API Name	Third-Party Device Execution Notification API				
Description	This API can be called to notify of the completion of third-party device execution. It is corresponding to the ***Request for Third-Party Device***. Repeated calls will be processed based on idempotence principle. Example: 1. RCS-2000 requests for elevator resources from WCS. 2. After elevator is in place, this API can be called to notify RCS-2000.				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/spi/wcs/robot/eqpt/notify				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message	Parameter	Data Type	Max. Length	Req. or Opt.	Description

Body	eqptCode	String	64	Req.	Third-party device No., which is a numeric string.
	taskCode	String	64	Req.	Task No., which is globally unique. UUID can be used.
	actionStatus	String	16	Req.	Task performance status. The enumeration can be extended. Predefined enumeration values: for automatic door and air shower gate: 1: door opening in position, 2: door closing in position, for elevator: 3: door opening in position, 4: material reaching the target floor, for loader/unloader, conveyor: 5: picking, 6: putaway, 7: arrival.
	siteCode	String	64	Req.	Allocated point No., which is used in scenes where multiple carriers are placed in an elevator.
	carrierInfo	Array of objects	N/A	Opt.	Carrier information, including carrier No. and layer No. (for CTU loader/unloader only).
	extra	JSON Object	N/A	Opt.	Custom extension field.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description

	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Dedicated Message Code	code		Description		
	Err_TaskNotStart		Task not started.		
	Err_TaskFinished		Task ended.		
	Err_TaskNotFound		Task not found.		
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
carrierInfo	carrierType	String	16	Opt.	Carrier type.
	carrierCode	String	64	Opt.	Carrier No.

Request Sample

```
POST / rcs/rtas/spi/wcs/robot/eqpt/notify HTTPS/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "eqptCode": "101",
    "taskCode": "180*****",
    "actionStatus": "1",
```

```
    "siteCode": "101",
    "extra": null
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "0",
    "message": "succeeded",
}
```

2.1.16 Apply Pre-Scheduling Task

API Name	Pre-Scheduling Task Applying API				
Description	This API can be called by the business system to apply pre-scheduling request, and RCS-2000 will call the AMR to the specified position in advance.				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/task/pretask				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				

Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	robotTaskCode	String	64	Opt.	Actual task No.
	siteCode	String	64	Req.	Point No. or point alias.
	nextTaskTime	String	64	Req.	Pre-scheduling time, indicating the estimated duration before an actual task is generated. Range: 0 to 3600, unit: seconds. The value 0 indicates clearing all pre-scheduling tasks of the point.
	robotType	String	N/A	Opt.	AMR type. Use comma (,) to separate multiple AMR types.
	priority	Integer	8	Opt.	Priority of the task, with larger value representing higher priority. The system will make a comprehensive assessment and dynamically adjust the actual priority based on the task busyness level, AMR working status, and other priority adjustment conditions. Range: 1 to 120.
	taskCount	Integer	N/A	Opt.	Number of pre-scheduled tasks (default: 1).
	capacityCount	Integer	N/A	Opt.	Min. number of empty storage bins of one AMR. It indicates the number of tasks (default: 1).
	amrDir	String	N/A	Opt.	Pre-scheduling target angle: 0, 90, 180, -90, 999.

					The value 999 indicates that the target angle is not specified.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	code	String	N/A	Req.	Message code.
	message	String	N/A	Req.	Message content.
	data robotTaskCode	String	64	Req.	Task No., which is globally unique.
	data extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	Err_TaskNotStart		Task not started.		
	Err_TaskFinished		Task ended.		
	Err_TaskNotFound		Task not found.		
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST / rcs/rtas/api/robot/controller/task/pretask HTTPS/1.1
Host: 10.10.10.10:1010
```

```
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "siteCode": "27e****ba08",
    "nextTaskTime": "60",
    "extra": null
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "robotTaskCode": "27e****ba08",
        "extra": null
    }
}
```

2.1.17 Search Task Status

API Name	Task Status Searching API
Description	This API can be called to search the task execution status by task order No.

Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/task/query				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	robotTaskCode	String	64	Req.	Task No., which is globally unique.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body	Field Name	Data Type	Max. Length	Req. or Opt.	Description

Business Data	robotTaskCode	String	64	Req.	Task No., which is globally unique.
	taskType	String	16	Req.	Task type. It is configured in RCS-2000, and it encapsulates internal scheduling logic that is independent of the business system. The enumeration can be extended. Predefined enumeration value: PF-LMR-COMMON: carrying, containing one or several execution steps.
	targetRoute	Array of objects	N/A	Req.	Collection of performance steps. After each step is completed, the business system is required to drive the next step via the task continuation API.
	initPriority	Integer	8	Opt.	Initial priority of the task, with higher value representing higher priority. The system will make a comprehensive assessment and dynamically adjust the actual priority based on the task busyness level, AMR working status, and other priority adjustment conditions.
	deadline	Time	N/A	Opt.	Task deadline, unit: sec. It is one of the task priority adjustment conditions. Task should be completed before the deadline.
	taskStatus	String	N/A	Req.	Task status. The enumeration values are fixed: QUEUE: in queue,

					WAIT: waiting for execution, EXECUTING: executing, MANUALED: manually completed, FINISHED: completed, CANCELLED: canceled.
	waring	JSON Object	1	Req.	Ongoing alarm. Only fault alarms under service alarms will be returned.
	singleRobotCode	String	32	Req.	The No. of the AMR that performs the task.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	Err_TaskCodeNotFound		Task does not exist.		
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	type	String	16	Req.	Target type. The enumeration can be extended. Predefined enumeration values: ZONE: target area No., SITE: point Location No., OPERAT: site No., STORAGE: bin No.
	code	String	64	Req.	The target No. corresponding to *type*.
	autoStart	Integer	1	Req.	Whether the step starts automatically. The enumeration values are fixed. Predefined enumeration values: 0: no,

					1: yes.
	operation	String	8	Opt.	Operation after AMR reaches the target position. The enumeration can be extended. Predefined enumeration values: COLLECT: picking, DELIVERY: putaway.
	robotType	String	N/A	Opt.	The AMR is selected from the specified range to perform the task. The business system ensures the correct AMR type used in the task. If the AMR selection range is specified manually for the current task, the scheduling system will ignore the range set in the task; if the AMR selection range is neither specified manually nor specified in the task, the scheduling system will look for the optimal solution within all available AMRs. Fixed enumeration values: GROUPS: AMR resource group No., ROBOTS: AMR No.
	robotCode	Array of strings	N/A	Opt.	The unique ID of the resource type that matches with the *robotType*.
	extra	JSON Object	N/A	Opt.	Custom extension field of *targetRoute*.
waring	taskWarnCode	String	16	Req.	Abnormal task alarm No.
	startTime	Time	N/A	Req.	Time of initial fault.
	singleRobotCode	String	16	Opt.	The unique ID of the AMR that is performing the task.
	errorCode	String	32	Req.	Custom fault code.

	errorMsg	String	256	Opt.	Custom fault message.

Request Sample

```
POST / rcs/rtas/api/robot/controller/task/query HTTPS/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "robotTaskCode": "27e****ba08",
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "robotTaskCode": "27e****ba08",
        "taskType": "PF-LMR-COMMON",
        "targetRoute": [
            {
                "type": "ZONE",
                "code": "7256****7a12",
```

```
                "autoStart": 1,
                "operation": "COLLECT"
            },{
                "type": "SITE",
                "code": "c9a1****3a1c",
                "autoStart": 0,
                "operation": "DELIVERY"
            }
        ],
        "initPriority": 53,
        "deadline": "2021-04-04T12:23:55Z",
        "taskStatus": "WAIT",
        "singleRobotCode": "f81***653",
        "currentSeq": 1,
        "waring": {
            "taskWarnCode": "20210404UNICODE9",
            "singleRobotCode": "f81***653",
            "startTime": "2021-04-04T12:23:55Z",
            "errorCode": "ABCDXYZ-12345678",
            "errorMsg": "Obstacle ahead",
        },
        "extra": null
    }
}
```


2.1.18 Search AMR Status

API Name	AMR Status Searching API
Description	This API can be called to search the status of an AMR by the AMR No.
Request	
Provider	RCS-2000
Service Prefix	/rcs/rtas
Path	/api/robot/controller/robot/query

Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	singleRobotCode	String	16	Req.	The unique ID of an AMR.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	singleRobotCode	String	16	Req.	AMR No.
	robotDir	Integer	32	Req.	AMR direction, range: [0, 360).
	robotIp	String	16	Opt.	AMR IP address.
	battery	Integer	32	Req.	AMR battery, range: [0, 100].
	x	String	32	Req.	X-coordinate of AMR current position.
	y	String	32	Req.	Y-coordinate of AMR current position.
	speed	Integer	32	Req.	AMR current speed, unit: mm/s.

	robotStatus	JSON Object	N/A	Req.	AMR current status.
	carrierCode	String	16	Opt.	The No. of the carrier on the AMR.
	warnings	Array of JSON Objects	N/A	Opt.	List of ongoing alarms. Only fault alarms under device alarms will be returned.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	N/A				
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
robotStatus	abnormal	String	N/A	Req.	Abnormal status. The enumeration values are fixed. YES: yes, NO: no.
	charging	String	N/A	Req.	Charging status. The enumeration values are fixed. YES: yes, NO: no.
	network	String	N/A	Req.	Network status. The enumeration values are fixed. ONLINE: online, OFFLINE: offline. Under offline status, information of other fields such as battery level and coordinates is inaccurate, so these fields can be ignored.
	taskable	String	N/A	Req.	AMR status, which is the same as that in AMR status map.

					The enumeration values are fixed. IDLE: idle, WORKING: executing task, PAUSE: paused.
	manual	String	N/A	Req.	Manual status. The enumeration values are fixed. MANUAL: manual, AUTO: automatic.
	emergency	String	N/A	Req.	Emergency stop status. The enumeration values are fixed. EMERGENCY: emergency stop, NORMAL: normal.
	extra	JSON Object	N/A	Opt.	Custom extension field.
warnings	taskWarnCode	String	16	Req.	Abnormal task alarm No.
	singleRobotCode	String	16	Opt.	The unique ID of the AMR.
	startTime	Time	N/A	Req.	Time of initial fault.
	errorCode	String	32	Req.	Custom fault code.
	errorMsg	String	256	Opt.	Custom fault message.

Request Sample

```
POST / rcs/rtas/api/robot/controller/robot/query HTTPS/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
```

```
    "singleRobotCode": "27e****ba08",
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "singleRobotCode": "f81***653",
        "robotDir": 180,
        "robotIp": "10.10.53.123",
        "battery": 56,
        "x": "1010.123",
        "y": "2301.225678",
        "speed": 1200,S
        "robotStatus": {
            "abnormal": "NO",
            "charging": "YES",
            "network": "ONLINE",
            "taskable": "IDLE",
            "manual": "AUTO",
            "emergency": "NORMAL",
            "extra": null
        },
        "waring": [
```

```
        {
            "taskWarnCode": "20210404UNICODE9",
            "singleRobotCode": "f81***653",
            "startTime": "2021-04-04T12:23:55Z",
            "errorCode": "ABCDXYZ-12345678",
            "errorMsg": "Obstacle ahead",
            "extra": null
        }
    ],
    "extra": {
        "actuatorInfo": {// CTU execution structure information
            "height": "1500",
            "carrierCode": "A001"
        },
        "binList": {//CTU/CMR storage bin information
            "row": "3",
            "column": "4",
            "level": "2",
            "bin": [
                {
                    "rowCount": "1",
                    "colCount": "1",
                    "levelCount": "1",
                    "carrierCode": "BIN001"
                },
                {
                    "rowCount": "2",
                    "colCount": "3",
                    "levelCount": "2",
                    "carrierCode": "BIN002"
                }
            ]
        }
    }
```

```
    }
}
```

2.1.19 Search Carrier Status

API Name	Carrier Status Searching API				
Description	This API can be called to search carrier current status. It also supports searching bin No. and AMR No. in versions later than 4.3.4.				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/carrier/query				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	carrierCode	String	16	Req.	Carrier No.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description

	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	carrierCode	String	16	Req.	Carrier No.
	robotTaskCode	String	64	Opt.	Current task No. of the carrier. It is empty if no tasks assigned.
	siteCode	String	32	Opt.	Point No. linked to the carrier. It is empty if no points linked.
	x	String	32	Opt.	X-coordinate of the carrier, which will be returned if the carrier is linked to a point.
	y	String	32	Opt.	Y-coordinate of the carrier, which will be returned if the carrier is linked to a point.
	carrierDir	Integer	32	Opt.	Rack direction relative to the point, range: [0, 360).
	carrierStatus	Integer	4	Req.	Carrier status. The enumeration values are fixed: NORMAL: normal, LOCKED: disabled.
	binCode	String	31	Opt.	Bin No., which will be returned if a carrier is on the bin.
	binName	String	64	Opt.	Customer bin No., which will be returned if a carrier is on the bin.
	robotCode	String	32	Opt.	The No. of an AMR. CTU No. will be returned if a carrier is on the CTU.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	N/A				
JSON Object Structure					

Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /rcs/rtas/api/robot/controller/carrier/query HTTPS/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "carrierCode": "27e****ba08",
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "carrierCode": "c51***3e8",
        "robotTaskCode": "27e****ba08",
        "siteCode": "ba46****ec31",
```

```
            "x": "2048.1024",
            "y": "1024.9854",
            "carrierDir": 270,
            "carrierStatus": "NORMAL",
            "binCode": "",
            "binName": "",
            "robotCode": "",
            "extra": null
        }
}
```


2.1.20 Link Carrier to Material

API Name	Carrier Linkage to Materials API				
Description	This API can be called only to link the idle rack to materials.				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/matlabel/bind				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
	Parameter	Data Type	Max. Length	Req. or Opt.	Description

Request Message Body	carrierCode	String	32	Req.	Carrier No.
	matLabel	String	64	Req.	Material label.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	code	String	N/A	Req.	Message code.
	message	String	N/A	Req.	Message content.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	N/A				
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /rcs/rtas/api/robot/controller/matlabel/bind HTTPS/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "carrierCode": "P00001",
    "matLabel": "M000****1"
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded"
}
```

2.1.21 Unlink Carrier from Material

API Name	Carrier Unlinking from Materials				
Description	This API can be called to unlink a carrier from materials.				
Request					
Provider	RCS-2000				
Service Prefix	/rcs/rtas				
Path	/api/robot/controller/matlabel/unbind				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	carrierCode	String	32	Req.	Carrier No.
	matLabel	String	64	Req.	Material label.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	code	String	N/A	Req.	Message code.

Business Data	message	String	N/A	Req.	Message content.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	N/A				
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /rcs/rtas/api/robot/controller/matLabel/unbind HTTPS/1.1

Host: 10.10.10.10:1010

Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"

X-lr-appkey: a4f********b324

X-lr-version: v1.0

X-lr-trace-id: 605****8a0b

X-lr-request-id: 393****a6c1

Content-Type: application/json;charset=UTF-8


{
    "carrierCode": "P00001",
    "matLabel": "M0000****1"
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded"
}
```

2.1.22 Notify of Third-Party Device Execution (Return Code: SUCCESS)

API Name	Third-Party Device Execution Notification API
Description	This API can be called to notify of the completion of third-party device execution. It is corresponding to the ***Request for Third-Party Device***. Repeated calls will be processed based on idempotence principle. Example: 1. RCS-2000 requests for elevator resources from WCS. 2. After elevator is in place, this API can be called to notify RCS-2000. The API functions are the same as those in ***Notify of Third-Party Device Execution (Return Code: 0)***, except that the return code is changed to "SUCCESS" in versions later than 4.2.8.
Request	
Provider	RCS-2000
Service Prefix	/rcs/rtas
Path	/spi/wcs/robot/eqpt/notifyGbt
Idempotence	Yes

Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	eqptCode	String	64	Req.	Third-party device No., which is a numeric string.
	taskCode	String	64	Req.	Task No., which is globally unique. UUID can be used.
	actionStatus	String	16	Req.	Task performance status. The enumeration can be extended. Predefined enumeration values: **for automatic door and air shower gate:** 1: door opening in position, 2: door closing in position, **for elevator:** 3: door opening in position, 4: material reaching the target floor, **for loader/unloader, conveyor:** 5: picking, 6: putaway, 7: arrival.
	siteCode	String	64	Req.	Allocated point No., which is used in scenes where multiple carriers are placed in an elevator.

	carrierInfo	Array of objects	N/A	Opt.	Carrier information, including carrier No. and layer No. (for CTU loader/unloader only).
	extra	JSON Object	N/A	Opt.	Custom extension field.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	taskCode	String	64	Req.	Task No., which is globally unique.
	extra	JSON Object	N/A	Opt	Custom extension field.
Dedicated Message Code	code		Description		
	Err_ TaskNotStart		Task not started.		
	Err_ TaskFinished		Task ended.		
	Err_ TaskNotFound		Task not found.		
JSON Object Structure					
Object Type	Data Type	Data Type	Max. Length	Req. or Opt.	Description
carrierInfo	carrierType	String	16	Opt	Carrier type.
	carrierCode	String	64	Opt	Carrier No.

Request Sample

```
POST /rcs/rtas/spi/wcs/robot/eqpt/notify HTTPS/1.1

Host: 10.10.10.10:1010

Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"

X-lr-appkey: a4f********b324

X-lr-version: v1.0

X-lr-trace-id: 605****8a0b

X-lr-request-id: 393****a6c1

Content-Type: application/json;charset=UTF-8


{
    "eqptCode": "101",
    "taskCode": "180*****",
    "actionStatus": "1",
    "siteCode": "101",
    "extra": null
}
```

Response Sample

```
HTTPS/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "robotTaskCode": "27e****ba08",
        "nextSeq": 1,
        "extra": null
    }
}
```

2.2 SPI APIs for Receiving Feedback

2.2.1 Receive Feedback on Task Performance

API Name	Task Performance Process Feedback API
Description	This API can be called to receive messages related to task performance process, including task starting, removing carrier from storage location, and task completion.
Request	
Provider	Upper-Level System
Service Prefix	
Path	/api/robot/reporter/task

Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	robotTaskCode	String	64	Req.	Task No.
	singleRobotCode	String	16	Req.	The unique ID of the AMR performing the task.
	extra	JSON Object	N/A	Opt.	Custom extension field.
	values	JSON Object	N/A	Opt.	
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	robotTaskCode	String	64	Req.	Task No., which is globally unique.
Dedicated Message Code	code		Description		
	N/A				

JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
values	mapCode	String	16	Req.	Map No.
	method	String	16	Req.	Method name for reporting messages related to task performance process. Default usage: start: task started, outbin: removing carrier from storage location, end: task completed.
	carrierCode	String	16	Req.	Carrier No.
	carrierName	String	256	Req.	Carrier name.
	carrierType	String	16	Req.	Carrier type.
	carrierCategory	String	16	Req.	Carrier category.
	carrierDir	String	4	Opt.	Carrier angle.
	slotCode	String	32	Opt.	Current point No.
	slotName	String	32	Opt.	Point alias. 1. For the status of getting out of storage location, it indicates the start point. 2. For the status of task completed, it indicates the target point.
	slotCategory	String	32	Opt.	Storage type. The enumeration values: BIN: bin, SITE: point.
	x	String	32	Opt.	X-coordinate of AMR current position.

	y	String	32	Opt.	Y-coordinate of AMR current position.
	amrCategory	String	32	Opt.	AMR category.
	amrType	String	32	Opt.	AMR type.
	amrCode	String	64	Opt.	AMR No., same as *single RobotCode*.
	zoneCode	String	16	Opt.	Area No.
	layerNo	Digits	N/A	Opt.	Stacking layer No. It indicates the layer in which the carrier is located.
	carrierWeight	String	16	Opt.	Carrier weight, unit: g (used for CTU only).

Request Sample

```
POST /api/robot/reporter/task HTTP/1.1

Host: 10.10.10.10:1010

Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"

X-lr-appkey: 542a*****8b15

X-lr-version: v1.0

X-lr-trace-id: 605****8a0b

X-lr-request-id: 393****a6c1

Content-Type: application/json;charset=UTF-8


{
    "robotTaskCode": "27e****ba08",
    "singleRobotCode": "8801",
    "extra": {
            "async": "0",
            "values": {
```

```
                "mapCode": "RR",
                "slotCategory": "SITE",
                "slotCode": "0146975RR0088902",
                "slotName": "0146975RR0088902",
                "x": 146975.0,
                "y": 88902.0,
                "method": "notifyPodArr",
                "carrierCategory": "POD",
                "carrierType": "R7",
                "carrierCode": "99013",
                "amrCategory": "LMR",
                "amrType": "5",
                "amrCode": "8801",
                "carrierName": "99013",
                "carrierDir": "-90.0",
                "layerNo": 1
            }
        }

}
```


Response Sample

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
```

```
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "robotTaskCode": "27e****ba08",
    }
}
```

2.2.2 Request for Resources

API Name	Resource Requesting API				
Description	This API can be called by RCS-2000 to request for resources from business system. For example, request for a rack target point from WMS.				
Request					
Provider	Upper-Level System				
Service Prefix					
Path	/api/robot/reporter/resource				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	robotTaskCode	String	64	Req.	Task No., which is globally unique.
	applyType	String	64	Req.	Request type. The enumeration can be extended. Predefined enumeration values:

					APPLY_SITE: apply for storage location, APPLY_BIN: apply for bin, APPLY_PTL_BIN: apply for PTL bin, APPLY_CARRIER: apply for carrier. APPLY_DERIVE_BIN: apply for bin to move carrier (used for iWMS-1000). APPLY_CTU_TRANS_AUTO_BIN: CTU transferring outbound (used for iWMS-1000).
	resourceType	String	64	Req.	Resource type. The enumeration can be extended. Predefined enumeration values: CARRIER: carrier, SITE: point. ZONE: area.
	resourceCode	String	64	Req.	The No. corresponding to *type*.
	label	String	64	Opt.	Correlation label. The correlation of applied resources is determined by the upper level system. The workstation No. will be input (used for iWMS-1000).
	relationCount	String	32	Opt.	Number of correlated resources.
	amrCategory	String	32	Opt.	AMR category.
	amrType	String	32	Opt.	AMR type.
	amrCode	String	16	Opt.	AMR No.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Response					

Status Code	Description					
200	The request succeeded.					
Response Header Field	Field Name		Data Type	Max. Length	Req. or Opt.	Description
	N/A					
Response Message Body Business Data	Field Name		Data Type	Max. Length	Req. or Opt.	Description
	data		JSON object	N/A	Req.	Custom extension field.
		type	String	16	Req.	Target type. The enumeration can be extended. Predefined enumeration values: ZONE: target area No., SITE: point No., STORAGE: bin No., CARRIER: carrier, CARRIER_TYPE: carrier type, MIX_CONDITION: mixed conditions.
		code	String	32	Req.	The target No. corresponding to *type*. For ZONE enumeration, it supports enumerating multiple areas separated by commas (,), for example, ZONE1, ZONE2. For MIX_CONDITION enumeration, the format is as follows: [{"type":"ZONE","code":"SF2"},{"type":" CARRIER_TYPE","code":"P1"}]
Dedicated Message Code	code			Description		
	N/A					

JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
extra	groupCode	String	32	Opt.	Group No.

Request Sample

```
POST /api/robot/reporter/resource HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "singleRobotCode": "f81***653",
    "applyType": "APPLY_SITE ",
    "resourceType": "SITE",
    "resourceCode": "27****08",
    "lable": "27****08",
    "extra": null
}
```

Response Sample

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
```

```
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "type": "SITE",
        "code": "27****08"
    }
}
```


2.2.3 Request for Third-Party Device

API Name	Third-Party Device Requesting API				
Description	This API can be called by RCS-2000 to request for third-party device control from third-party device service, for example, apply for elevator resources from WCS.				
Request					
Provider	Upper-Level System				
Service Prefix					
Path	/api/robot/reporter/eqpt				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	eqptCode	String	64	Req.	Third-party device No., which is a numeric string.
	eqptName	String	64	Req.	Third-party device name.

	taskCode	String	64	Req.	Task No., which is globally unique. UUID can be used.
	method	String	64	Req.	Task execution method, which is device dependent. The enumeration can be extended. Predefined enumeration values: CANCEL: cancel task, APPLY_TO_AGV: apply for picking by AMR, APPLY_FROM_AGV: apply for putaway by AMR, ARRIVED: AMR arrives, RELEASE: AMR leaves. Predefined enumerations for automatic doors: APPLY_LOCK: apply for automatic door opening, RELEASE_EQPT: release automatic door. Predefined enumerations for elevators: APPLY_RESOURCE: apply for elevators, EXECUTE_TASK: elevator arrives at target floor. RELEASE_RESOURCE: release elevators.
	carrierInfo	Array of objects	N/A	Opt.	Carrier information, including carrier No. and layer No.
	srcCode	String	64	Opt.	Start position No. If the start position is an elevator, the start floor No. should be adopted. For air shower gates: 1: front gate, 2: rear gate.

	dstCode	String	64	Opt.	Target position No. If the target position is an elevator, the target floor No. should be adopted.
	srcPathDir	Integer	32	Opt.	Passing direction through the automatic door. 1: [-180, 0), 2: (0, 180]. For elevator doors: 1: front door, 2: rear door.
	dstPathDir	Integer	32	Opt.	For elevator doors: 1: front door, 2: rear door. TAS gets the elevator door side based on the waiting point.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	N/A				
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description

carrierInfo	carrierType	String	16	Opt.	Carrier type.
	carrierCode	String	64	Opt.	Carrier No.
	layer	String	4	Opt.	Layer No., bottom-up numbering starting from 0.

Request Sample

```
POST /api/robot/reporter/eqpt HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "eqptCode": "101",
    "taskCode": "27e****ba08",
    "method": "APPLY_TO_AGV"
}
```

Response Sample

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
```

```
            "extra": null
        }
}
```

2.2.4 Receive Feedback on Completion of AMR Returning

API Name	AMR Returning Feedback API				
Description	This API can be called by the business system to receive the feedback on completion of AMR returning. You can log in to RCS-2000, go to Settings > Extension > Data Sync Permission > Service Notification, and enable ZONE_HOMING to enable the API.				
Request					
Provider	Upper-Level System				
Service Prefix					
Path	/api/robot/reporter/zone/homing				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	homingCode	String	32	Req.	Returning command No.
	workRobotCode	Array of strings	N/A	Opt.	The No. array of AMRs which are still working.
	status		N/A	Req.	AMR returning task execution status.
Response					

Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	N/A				
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /api/robot/reporter/zone/homing HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: b4f********f324
X-lr-version: v1.0
X-lr-trace-id: 665****8a0b
X-lr-request-id: 363****a6c1
Content-Type: application/json;charset=UTF-8

{
"homingCode": "821****8ac7",
"status": "SUCCESS"
}
```

Response Sample

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "extra": null
    }
}
```

2.2.5 Receive Feedback on Completion of Area Clearing

API Name	Area Clearing Feedback API				
Description	This API can be called by the business system to receive the feedback after all AMRs return to the fixed area or not all returning tasks complete when time is up. You can log in to RCS-2000, go to Settings > Extension > Data Sync Permission > Service Notification, and enable ZONE_BANISH to enable the API.				
Request					
Provider	Upper-Level System				
Service Prefix					
Path	/api/robot/reporter/zone/banish				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				

Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	banishCode	String	32	Req.	Area clearing command No.
	areaList	Array of objects	N/A	Opt.	Area list.
	stayRobotCode	Array of strings	N/A	Opt.	The No. array of AMRs still in the area.
	status	String	64	Opt.	AMR clearing task execution status. SUCCESS: all tasks succeed, FAIL: not all tasks succeed when time is up.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	N/A				
JSON Object Structure					

Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
areaList	areaCode	String	32	Req.	Area No.

Request Sample

```
POST /api/robot/reporter/zone/banish HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
"banishCode": "27e****ba08",
"status": "SUCCESS"
}
```

Response Sample

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "extra": null
    }
```

```
}
```

2.2.6 Report AMR Exception Alarm

API Name	AMR Exception Alarm Reporting API				
Description	This API can be called by RCS-2000 to report the severe alarm which causes AMR stopping to the upper level system. The alarm will be reported only once. You can log in to RCS-2000, go to Settings > Extension > Data Sync Permission > Service Notification, and enable AMR_ALARM to enable the API.				
Request					
Provider	Upper-Level System				
Service Prefix					
Path	/api/robot/reporter/robot/warning				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	singleRobotCode	String	16	Req.	The unique ID of the abnormal AMR.
	taskWarnCode	String	16	Req.	Abnormal task alarm No.
	startTime	Time	N/A	Req.	Time of initial fault.
	robotTaskCode	String	64	Opt.	The task No. that the AMR is performing.

	x	String	32	Req.	X-coordinate of AMR current position.
	y	String	32	Req.	Y-coordinate of AMR current position.
	errorCode	String	32	Req.	Custom fault code.
	errorMsg	String	256	Opt.	Custom fault message.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	N/A				
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /api/robot/reporter/robot/warning HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
```

```
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "taskWarnCode": "20210404UNICODE9",
    "singleRobotCode": "f81***653",
    "startTime": "2021-04-04T12:23:55Z",
    "x": "1010.123",
    "y": "2301.225678",
    "errorCode": "ABCDXYZ-12345678",
    "errorMsg": "Obstacle ahead",
    "extra": null
}
```

Response Sample

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "extra": null
    }
}
```

2.2.7 Report Task Exception Alarm

API Name	Task Exception Alarm Reporting API				
Description	This API can be called by RCS-2000 to report the severe alarm during the task to the upper level system. The alarm will be reported only once. You can log in to RCS-2000, go to Settings > Extension > Data Sync Permission > Service Notification, and enable TASK_ALARM to enable the API.				
Request					
Provider	Upper-Level System				
Service Prefix					
Path	/api/robot/reporter/task/warning				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	robotTaskCode	String	64	Req.	The unique ID of the abnormal task.
	taskWarnCode	String	16	Req.	Abnormal task alarm No.
	mapCode	String	16	Opt.	The map No.
	startTime	Time	N/A	Req.	Time of initial fault.
	singleRobotCode	String	16	Opt.	The unique ID of the AMR that is performing the task.
	errorCode	String	32	Req.	Custom fault code.

	errorMsg	String	256	Opt.	Custom fault message.
	extra	JSON Object	N/A	Opt.	Custom extension field.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	N/A				
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /api/robot/reporter/task/warning HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8
```

```
{
    "taskWarnCode": "20210404UNICODE9",
    "robotTaskCode": "27e****ba08",
    "singleRobotCode": "f81***653",
    "startTime": "2021-04-04T12:23:55Z",
    "errorCode": "ABCDXYZ-12345678",
    "errorMsg": "Obstacle ahead",
    "extra": null
}
```

Response Sample

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "SUCCESS",
    "message": "succeeded",
    "data": {
        "extra": null
    }
}
```

2.2.8 Notify of Linking/Unlinking

API Name	Third-Party Linking/Unlinking Notification API
Description	This API can be called by RCS-2000 to report the linking/unlinking message to the upper level system. The corresponding upper level system can be configured in the service notification. You can log in to RCS-2000, go to Settings > Extension > Data Sync Permission > Service Notification, and enable BIND to enable the API.

Request					
Provider	Upper-Level System				
Service Prefix					
Path	/api/robot/reporter/bind				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	slotCategory	String	16	Req.	The type of storage objects. SITE: point, BIN: bin.
	slotCode	String	64	Req.	The storage object No.
	carrierCategory	String	16	Req.	The carrier category. POD: rack, PALLET: pallet, BOX: tote, MAT: material.
	carrierType	String	16	Opt.	The carrier type.
	carrierCode	String	16	Opt.	The carrier No.
	carrierDir	Integer	N/A	Opt.	The carrier direction, range: [-179, 360).

	carrierHeight	Integer	N/A	Opt.	The carrier height.
	temporary	Integer	N/A	Req.	Whether the carrier is temporary. 0: no, 1: yes.
	colCount	Integer	N/A	Req.	The layer where a carrier is located.
	invoke	String	16	Req.	The operation type. BIND: linking, UNBIND: unlinking.
	areaCode	String	32	Opt.	The area No. of a storage object.
	upperCarriers	Array of objects	N/A	Opt.	The carrier information, excluding the first layer of a carrier. It is used for unlinking the stacked carriers via point. And it is supported in the version later than V4.2.10.

Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated	code		Description		

Message Code	N/A				
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
upperCarriers	carrierCategory	String	16	Req.	The carrier category. POD: pallet, PALLET: pallet, BOX: tote, MAT: material.
	carrierType	String	16	Req.	The carrier type.
	carrierCode	String	16	Req.	The carrier No.
	colCount	Integer	N/A	Req.	The layer where a carrier is located.

Request Sample

```
POST /api/robot/reporter/bind HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "slotCategory": "SITE",
    "slotCode": "0553633LL0026800",
```

```
        "carrierCategory": "POD",
        "carrierType": "P3",
        "carrierCode": "P802",
        "carrierDir": 90,
        "carrierHeight": 2400,
        "temporary": 0,
        "colCount": 1,
        "invoke": "BIND",
       "areaCode": "PT02",
       "upperCarriers": [
              {
                  "carrierCategory": "PALLET",
                  "carrierCode": "SYS_1981D0063DB112T",
                  "carrierType": "22",
                  "colCount": 2
              }
          ]
}
```


Response Sample


```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

                                                    {
       "code": "SUCCESS",
        "message": "succeeded",
       "data": {
            "extra": null
       }
```

```
}
```

2.2.9 Verify Tote Inbound at FlashStation

API Name	FlashStation Tote Inbound Verification API				
Description	This API can be called by RCS-2000 to check if the tote No. is valid when the tote is conveyed to FlashStation. The upper level system calls the common API, and RCS-2000 calls the verification API of upper level system. If a success message is returned, the tote is verified. You can log in to RCS-2000, go to Settings > Extension > Data Sync Permission > Service Notification to enable the API (No.: WORKSTATION_CHECK_CARRIER).				
Request					
Provider	Upper-Level System				
Service Prefix					
Path	/api/robot/reporter/applyReturnForValid				
Idempotence	Yes				
Request Method	POST				
Search Parameter	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Request Message Body	Parameter	Data Type	Max. Length	Req. or Opt.	Description
	robotTaskCode	String	64	Req.	The task No.
	type	String	64	Req.	The verification resource type. CARRIER: carrier.

	code	String	64	Req.	The resource No.
	label	String	64	Opt.	The point No. of the current FlashStation.
Response					
Status Code	Description				
200	The request succeeded.				
Response Header Field	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	N/A				
Response Message Body Business Data	Field Name	Data Type	Max. Length	Req. or Opt.	Description
	extra	JSON Object	N/A	Opt.	Custom extension field.
Dedicated Message Code	code		Description		
	N/A				
JSON Object Structure					
Object Type	Parameter	Data Type	Max. Length	Req. or Opt.	Description
N/A					

Request Sample

```
POST /api/robot/reporter/applyReturnForValid HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8
```

```
{
    "robotTaskCode": "1********C",
    "type": "CARRIER",
    "code": "BOX00001",
    "label": "0010000AA0010000"
}
```


Response Sample


```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
X-lr-request-id: 393****a6c1
X-lr-trace-id: 605****8a0b
X-lr-version: v1.0
Date: Fri, 26 Mar 2021 06:46:18 GMT

{
    "code": "0",
"message": "succeeded"
}
```

Chapter 3 Samples for Calling Task Applying API

3.1 LMR General Transportation Task

```
POST /api/robot/controller/task/submit HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "taskType": "PF-LMR-COMMON",
     "targetRoute": [{
            "type": "CARRIER",
            "code": "jieb1"
     }, {
            "type": "SITE",
            "code": "****"
     }],
    "initPriority": 1
}
```

3.2 LMR Task – Initialization of Specified Carrier

The carrier should not be linked at the start point.

```
POST /api/robot/controller/task/submit HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
```

```
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "taskType": "PF-LMR-INIT-CARRIER",
     "targetRoute": [
        {
            "type": "SITE",
            "code": "XXXXXXX",
        },
        {
            "type": "ZONE",
            "code": "SL2",
              "extra": {
                  "angleInfo": {
                        "code": "90"
                  }
              }
        }
    ],
    "initPriority": 10,
    "extra": {
         "carrierInfo": [{
              "carrierType": "5",
              "carrierCode": "P00001"
```

```
            }]
        },
        "interrupt": 0
}
```

3.3 Carrier Detection and Transportation

```
POST /api/robot/controller/task/submit HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "taskType": "PF-DETECT-CARRIER",
    "targetRoute": [
        {
            "type": "SITE",
            "code": "****",
        },
        {
            "type": "SITE",
            "code": "*****",
        }
    ],
    "initPriority": 10,
    "extra": {
        "carrierInfo": [{
            "carrierType": "66"
```

```
            }]
        },
        "interrupt": 0
}
```

3.4 CTU General Transportation Task

```
POST /api/robot/controller/task/submit HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "taskType": "PF-CTU-COMMON",
    "targetRoute": [
        {
            "type": "CARRIER",
            "code": "C0001",
        },
        {
            "type": "STORAGE",
            "code": "***",
        }
    ],
    "initPriority": 10,
    "extra": {
        "carrierInfo": [{
            "carrierType": "66"
        }]
```

```
    },
    "interrupt": 0
}
```

3.5 CTU Workstation Inbound

```
POST /api/robot/controller/task/submit HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "taskType": "PF-CTU-LOADER-IN",
    "targetRoute": [{
        "type": "EQPT",
        "code": "222"
    }],
    "initPriority": 1,
    "extra": {
        "carrierInfo": [{
            "carrierType": "3",
            "carrierCode": "jieb1",
            "layer": 0
        }, {
            "carrierType": "3",
            "carrierCode": "jieb2",
            "layer": 1
        }, {
            "carrierType": "3",
```

```
                "carrierCode": "jieb3",
                "layer": 2
            }, {
                "carrierType": "3",
                "carrierCode": "jieb4",
                "layer": 3
            }]
        }
}
```


3.6 CTU Task – Buffer Rack Outbound

```
POST /api/robot/controller/task/submit HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "taskType": "WMS-PF-CTU-ALLOC-WALL-OUT",
    "targetRoute": [{
        "type": "CARRIER",
        "code": "jieb1"
    }, {
        "type": "PTL_WALL",
        "code": "900003"
    }],
    "initPriority": 1
}
```

3.7 CTU Task – Buffer Rack Inbound

```
POST /api/robot/controller/task/submit HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "taskType": "WMS-PF-CTU-ALLOC-WALL-IN",
    "targetRoute": [{
        "type": "STORAGE",
        "code": "R900003A03021"
    }],
    "extra": {
        "carrierInfo": [{
            "carrierType": "3",
            "carrierCode": "jieb1"
        }]
    }
    "initPriority": 1
}
```

3.8 CTU Conveyor Detection Inbound

```
POST /api/robot/controller/task/submit HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
```

```
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "taskType": "PF-CTU-CONVEYOR-IN-DETECT",
    "targetRoute": [{
        "type": "EQPT",
        "code": "777"
    }],
    "initPriority": 1
}
```

3.9 CTU Conveyor Inbound

```
POST /api/robot/controller/task/submit HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "taskType": "PF-CTU-CONVEYOR-IN",
    "targetRoute": [{
        "type": "EQPT",
        "code": "777"
    }],
    "initPriority": 1,
    "extra": {
        "carrierInfo": [{
```

```
                "carrierType": "3",
                "carrierCode": "901"
            }]
        }
}
```

3.10 FMR General Transportation Task

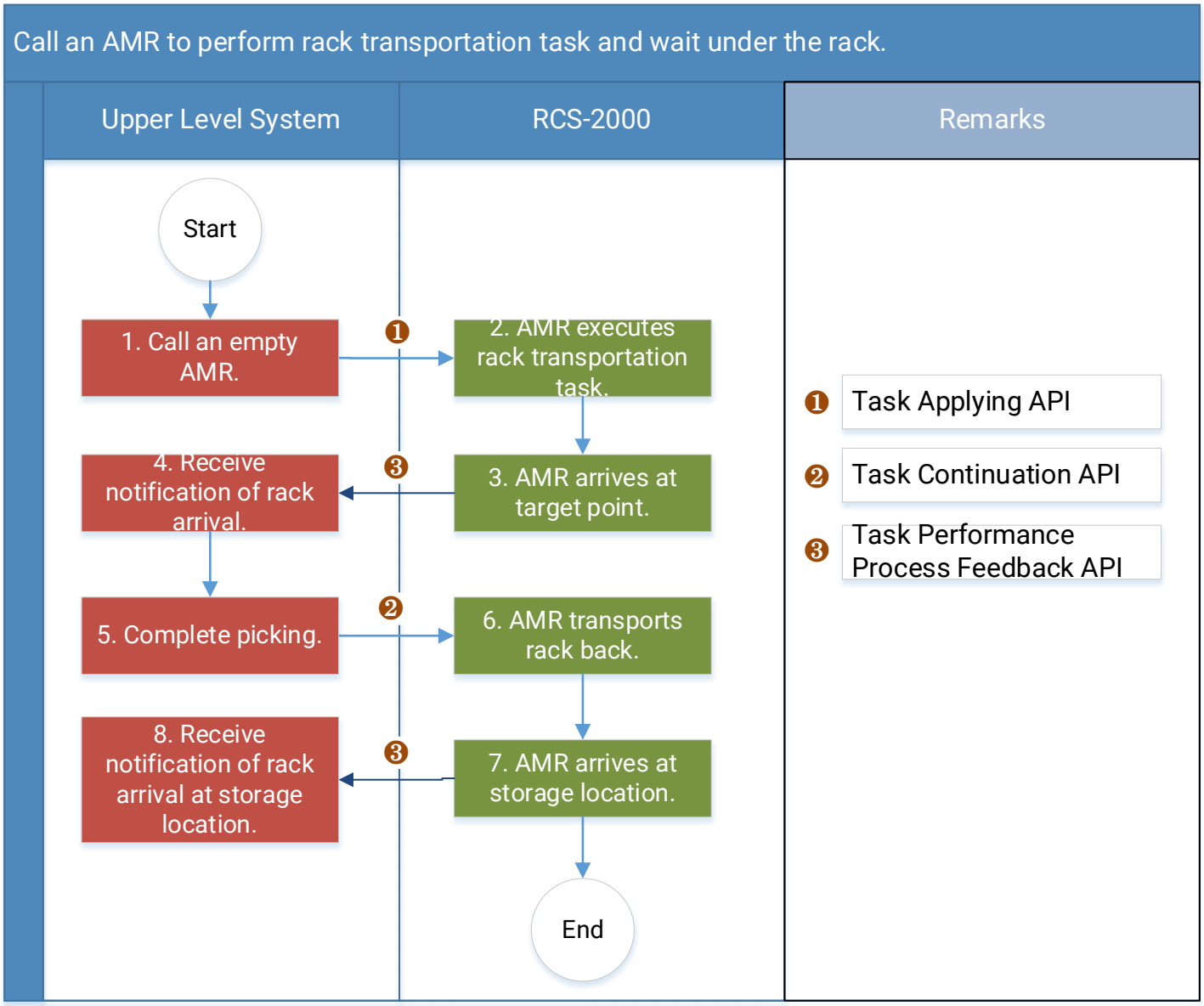
```
POST /api/robot/controller/task/submit HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "taskType": "PF-FMR-COMMON",
     "targetRoute": [
        {
            "type": "CARRIER",
            "code": "PA0001",
        },
        {
            "type": "ZONE",
            "code": "SL2",
        }
    ],
    "initPriority": 10,
     "interrupt": 0
}
```

3.11 Roller CMR General Transportation Task

```
POST /api/robot/controller/task/submit HTTP/1.1
Host: 10.10.10.10:1010
Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"
X-lr-appkey: a4f********b324
X-lr-version: v1.0
X-lr-trace-id: 605****8a0b
X-lr-request-id: 393****a6c1
Content-Type: application/json;charset=UTF-8

{
    "taskType": "PF-CMR-COMMON",
     "targetRoute": [
        {
            "type": "SITE",
            "code": "012***0512",
        },
        {
            "type": "SITE",
            "code": "022***0843",
        }
    ],
    "initPriority": 10,
     "interrupt": 0
}
```

Chapter 4 Typical Scheduling Scenes

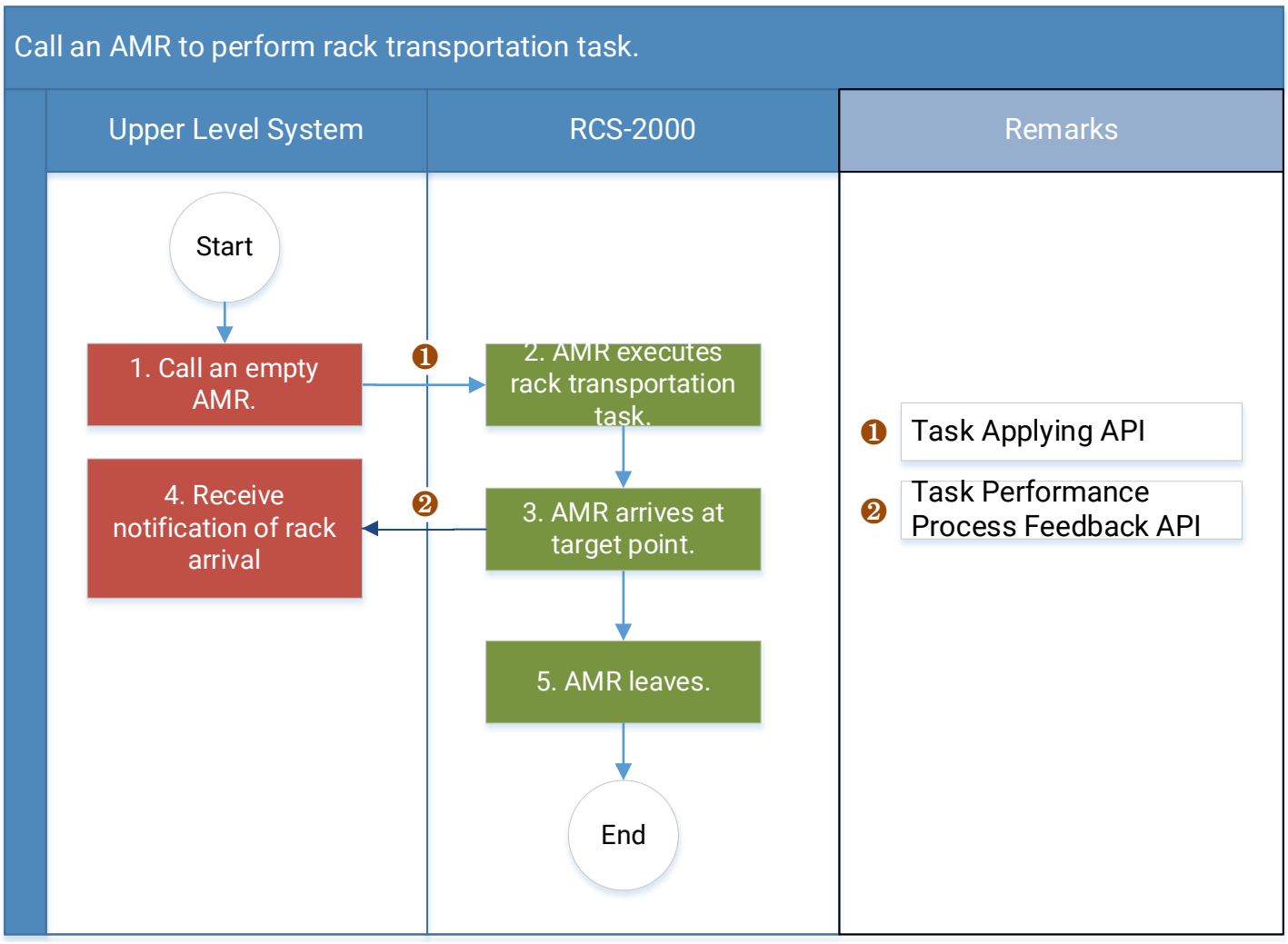
Scene 1: Call an AMR to perform rack transportation task and wait under the rack.



See Also

- Task Applying API
- Task Continuation API
- Task Performance Process Feedback API

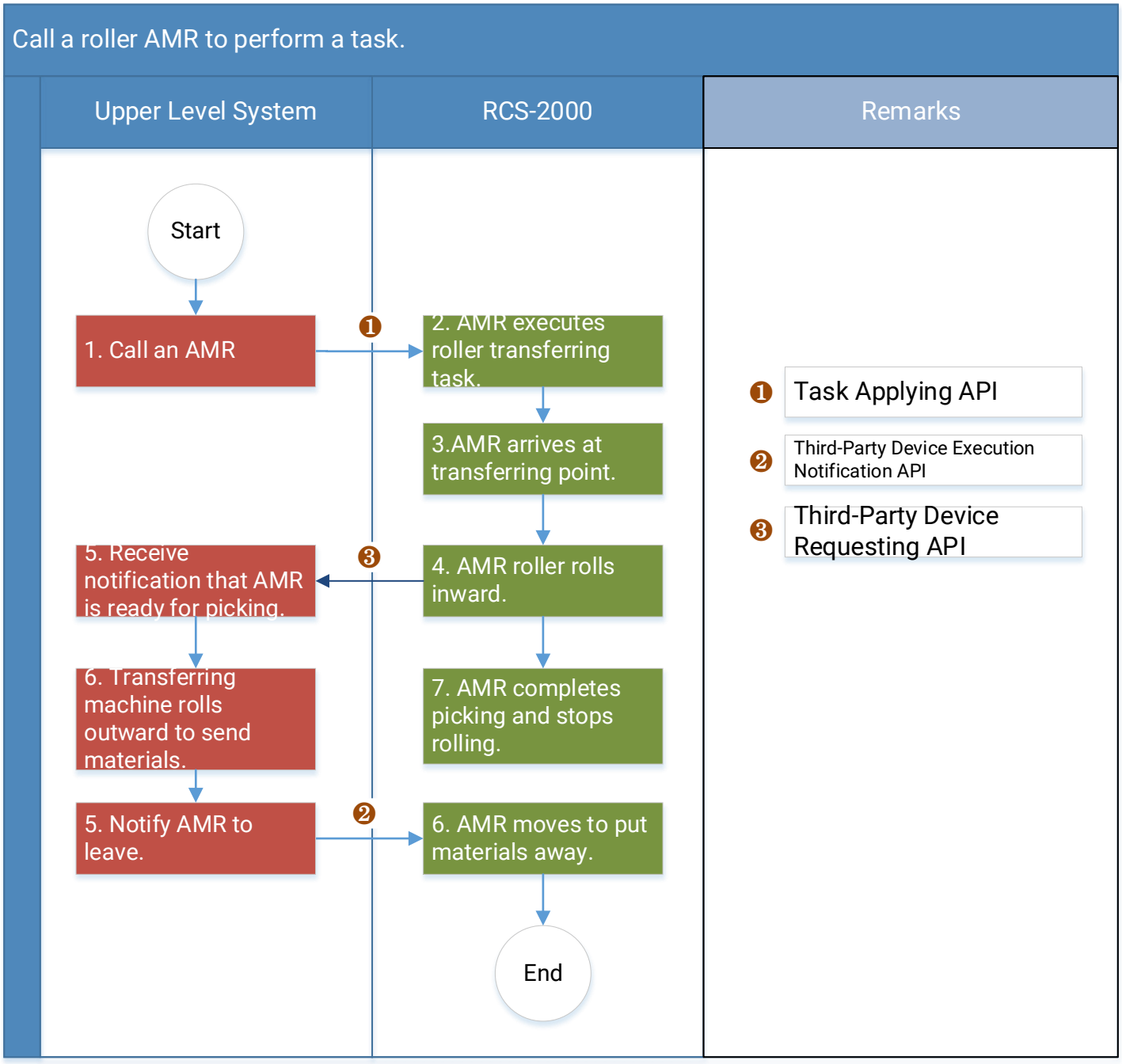
Scene 2: Call an AMR to perform rack transportation task.



See Also

- Task Applying API
- Task Performance Process Feedback API

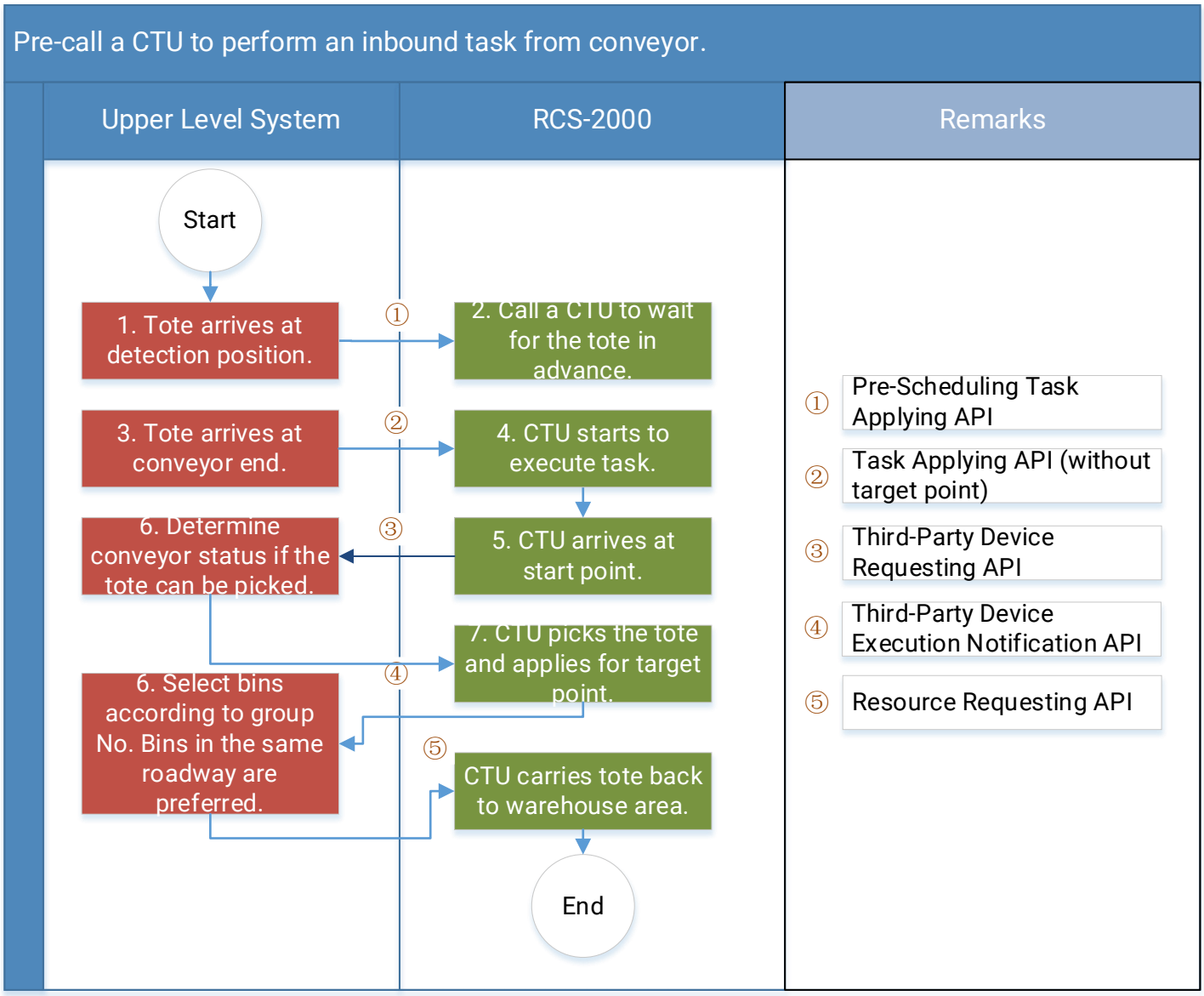
Scene 3: Call a roller AMR to perform a task.



See Also

- Task Applying API
- Third-Party Device Execution Notification API
- Third-Party Device Requesting API

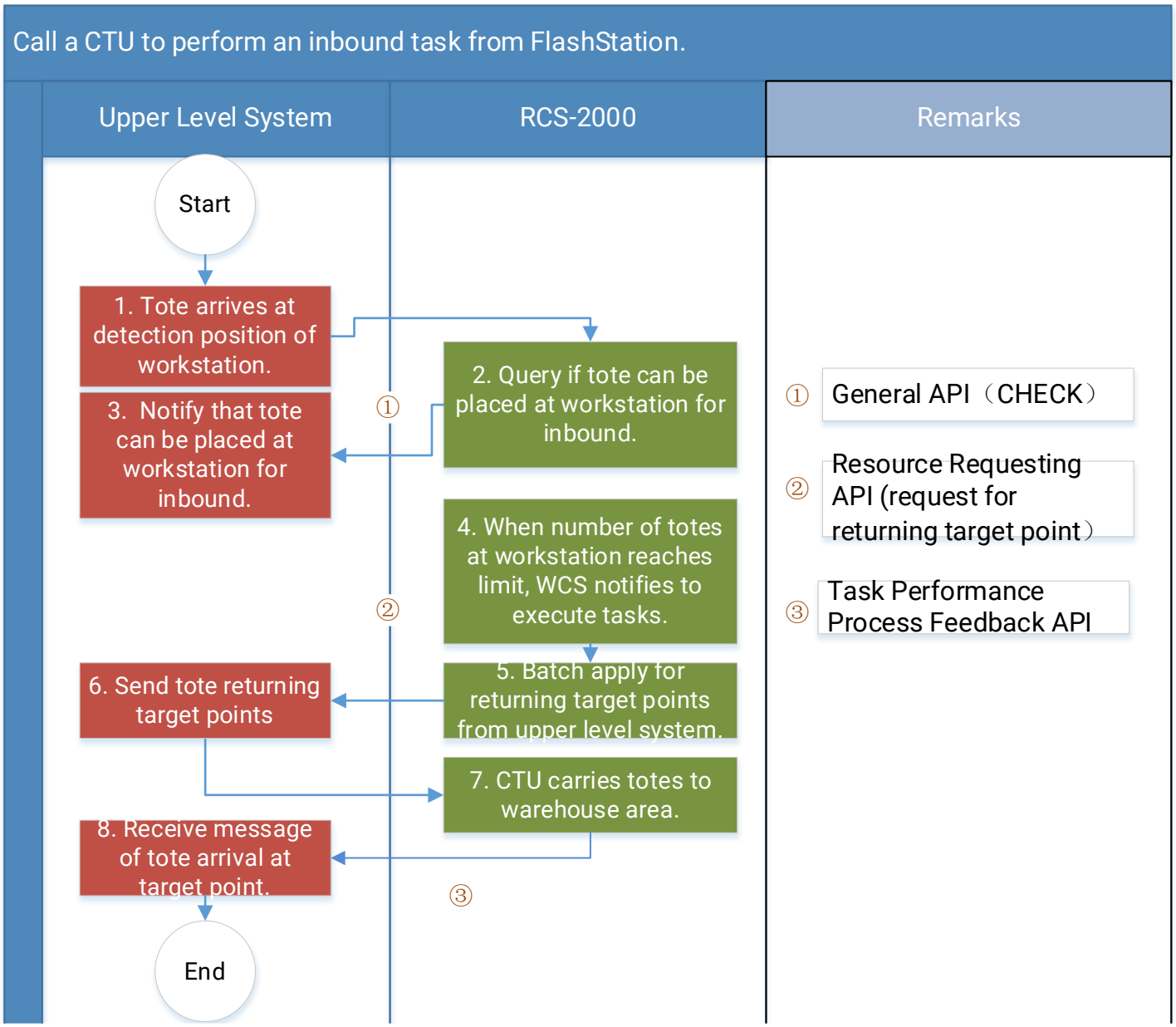
Scene 4: Pre-call a CTU to perform an inbound task from conveyor.



See Also

- Pre-Scheduling Task Applying API
- Task Applying API
- Third-Party Device Requesting API
- Third-Party Device Execution Notification API
- Resource Requesting API

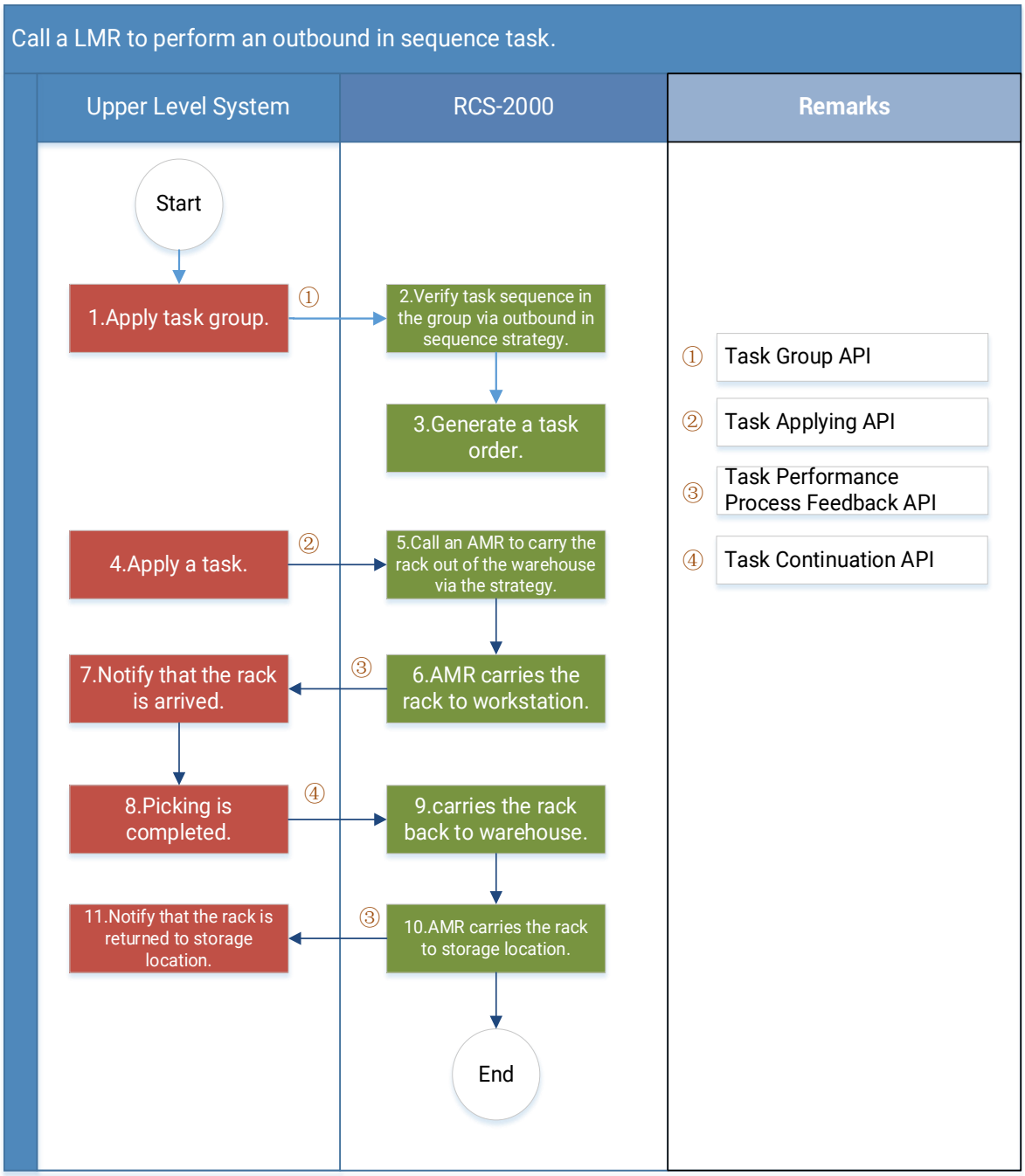
Scene 5: Call a CTU to perform an inbound task from FlashStation.



See Also

- General API
- Resource Requesting API
- Task Performance Process Feedback API

Scene 6: Call a LMR to perform an outbound-in-sequence task.



See Also

- Task Group API
- Task Performance Process Feedback API
- Task Continuation API

API calling methods for outbound in sequence:

Method 1 (Recommended)

Step 1 Call the Task Group API, select the outbound in sequence strategy, and generate a task order which is in queue.

Step 2 Call the Task Applying API, input the task group No. and task chain No. identified in Step 1 to task message, and apply the specified task.

Method 2

Step 1 Call the Task Applying API, and input the group sequence and outbound in sequence task group information to task message. RCS-2000 will generate a task group, and generate a task chain.

Step 2 Input the group No. in groupCode, and input the following parameters in extra:

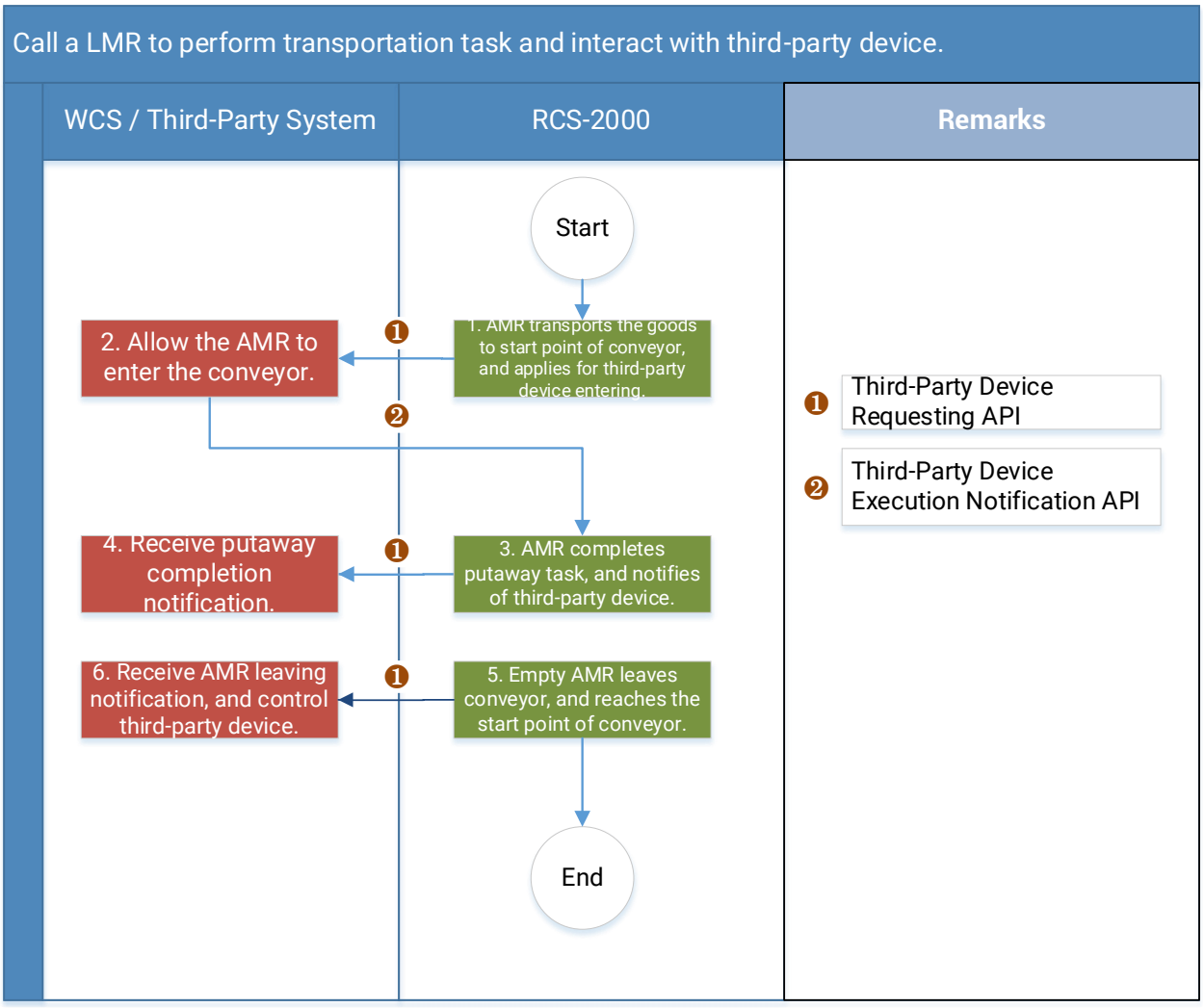
- seqType (Required): The outbound in sequence scene. It is corresponding the value of strategyValue in Task Group API. This parameter supports all outbound in sequence scenes in Task Group API, enumeration range: 0, 1, 2, and 3.
- *sequence* (Required): The sequence in the task.
- *groupId* (Optional): The task sequence.
- If the value of *groupId* is not input, *groupCode* should be parsed as task sequence. The value of *groupCode* should be an integer, otherwise, a parsing exception will be thrown out.

Request Sample

```
POST /rcs/rtas/api/robot/controller/task/submit HTTP/1.1

Host: 10.10.10.10:1010

Authorization: nonce="wab1tkh",method="HMAC-SHA256",timestamp="2021-01-01T00:00:00+08:00"

X-lr-appkey: a4f********b324

X-lr-version: v1.0

X-lr-trace-id: 605****8a0b

X-lr-request-id: 393****a6c1

Content-Type: application/json;charset=UTF-8


{

    "taskType": "PF-LMR-COMMON",

    "targetRoute": [

        {
```

```
                "type": "CARRIER",
                "code": "725",
                "operation": "COLLECT"
            },{
                "type": "SITE",
                "code": "c9a1****3a1c",
                "operation": "DELIVERY"
            }],
        "initPriority": 1,
        "groupCode": "22",
        "extra": {
            "sequence":1,
            "groupId": 1,
            "seqType":"1"
        }
}
```

Scene 7: Call a LMR to perform transportation task and interact with third-party device.



See Also

- Third-Party Device Requesting API
- Third-Party Device Execution Notification API

Other applications of APIs in CTU scenes:

- Inbound and outbound by using PTL: Task Applying API.
- Outbound of a single material or in group to conveyor or FlashStation: Task Applying API.
- Outbound in sequence to conveyor of FlashStation: First call the Task Group API, and then call the Task Applying API.
- Inbound by using conveyor: Task Applying API.
- Applying for tote picking and putaway, and notifying of task completion: Task Performance Process Feedback API.
- Upper-level system allowing CTU to pick or put away totes: Third-Party Device Execution Notification API.